

SELF-CONSISTENT STOCHASTIC INTERPOLANTS FOR COVARIANCE PRIOR AND POSTERIOR LEARNING

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ABSTRACT

We present a method for sampling a covariance matrix conditioned on its empirical estimate, using only an ensemble of empirical covariances, without access to the underlying covariance prior or clean training pairs. Conditional stochastic interpolants trained on reconstructed covariances and freshly simulated observations provide a practical implementation of expectation–maximization (EM): in the ideal population limit, posterior draws averaged over observations update the covariance prior. A general implementation evolves matrix-logarithm coordinates. Under rotational invariance, a scalable alternative learns only the posterior distribution of the population spectrum and completes each full covariance draw by sampling its relative orientation from a known conditional distribution. We establish a sharp sample-size threshold for identifiability of an unrestricted covariance prior, convergence of the ideal EM iteration in observation space, and prior and posterior recovery under explicit additional conditions. In the rotationally invariant case, exact orientation completion transfers spectral-posterior guarantees directly to the full covariance posterior. We also give conditional error bounds for imperfectly learned stochastic interpolants. We evaluate the method on synthetic examples with exact reference posteriors and on EEG covariance data, assessing both posterior reconstruction and downstream covariance estimation.

Working draft. Section 5 describes proposed experiments. No numerical performance or uncertainty-calibration results are claimed. Detailed proofs and assumptions are in the appendices.

1 INTRODUCTION

How can we sample a covariance matrix conditioned on its empirical estimate when neither the underlying covariance prior nor clean training examples are available? We consider repeated covariance-estimation problems: an unobserved matrix C is drawn from an unknown population distribution π_* , and C_e is the empirical covariance of N Gaussian observations with covariance C . We observe many C_e ’s, but no corresponding C ’s. We cannot evaluate or sample π_* . Our objective is nevertheless to approximate $P_*(C \mid C_e)$, representing uncertainty about the entire covariance by conditional draws rather than a single estimate.

Knowing how to simulate C_e from a supplied C does not solve this problem. The simulator cannot supply clean draws from the unknown population, and the likelihood alone does not determine a posterior. The empirical-covariance ensemble must support both learning the shared prior and constructing its posterior sampler. Even a current prior represented through generated samples need not have a tractable posterior. Thus an EM formula that assumes conditional samples is not yet a computational method.

We combine the conditional stochastic interpolants of Chen et al. (2024) with the reconstruction–re-corruption approach of Modi et al. (2026). A stochastic interpolant is a noisy path between an observation and a paired target. Squared-error regression learns the drift, or mean direction of change, of a stochastic differential equation (SDE) that samples the conditional target distribution. Self-consistency supplies the missing pairs: apply the current sampler to real observations, simulate fresh empirical covariances from its reconstructions, and train the next sampler on those synthetic pairs. A chosen, sampleable initialization starts the procedure; it is not access to the true prior.

We develop two implementations of this idea for the same full-covariance posterior objective. The general method learns an SDE in matrix-logarithm coordinates, ensuring positive-definite outputs. For a rotationally invariant (RI) prior, only the spectral conditional distribution must be learned. The remaining eigenvector distribution is explicitly known once the spectrum and observation are fixed. We sample it separately, after training, without another learned SDE. This reduces the dimension of the learned state without replacing full-matrix inference by spectral point estimation.

Related work. Conditional interpolants (Chen et al., 2024) provide observation-initialized SDEs and a post-training choice of noise coefficient, including the Föllmer process. Self-consistent stochastic interpolants (SCSI) (Modi et al., 2026) supply reconstruction and fresh corruption for learning from a forward simulator. Diffusion-based EM methods also learn clean distributions from corrupted observations (Rozet et al., 2024; Bai et al., 2024); DiffEM is particularly close, using conditional diffusion with reconstruction and re-corruption (Hosseintabar et al., 2025). We build on these methods rather than claim conditional EM as a new general principle. The distribution-level analysis of Aubin-Frankowski et al. (2022) supplies the relevant EM foundation. Our concern is making conditional covariance sampling explicit and establishing what the resulting ideal iteration identifies and recovers. The conditional bounds for fitted samplers are extensions of this foundation.

Linear (Ledoit & Wolf, 2004) and nonlinear (Ledoit & Wolf, 2012) covariance shrinkage adjust empirical eigenvalues for point estimation. Ledoit–Péché (LP) theory distinguishes population eigenvalues from variances along empirical eigenvectors (Ledoit & Péché, 2011). These results do not require an RI random prior; RI here is the assumption enabling an exact posterior reduction. Conditional orthogonal-matrix sampling connects to the matrix Bingham methods of Hoff (2009). We retain the full posterior as the target and extract loss-dependent covariance estimates afterward.

Contributions and organization. Section 2 poses the observed-data problem and ideal EM update. Section 3 supplies its missing posterior sampler through a matrix-logarithm SI and establishes identifiability and ideal population convergence, distinguishing observation-law convergence from prior and posterior recovery. Section 4 reduces spectral learning without changing the full-matrix objective or ideal guarantees. Both algorithms have adjustable interpolation noise. Scalar/diagonal examples and incomplete-learning bounds are in the appendices. Section 5 proposes tests of the theory and implementations, not completed results.

2 PROBLEM SETUP AND THE IDEAL EM ITERATION

2.1 OBSERVED ENSEMBLE AND THE KNOWN FORWARD MECHANISM

Given an (unknown) prior distribution $\pi_\star$ over the set $\mathbb{S}_{++}^d$ of positive-definite symmetric $d \times d$ matrices, let

$$C \sim \pi_\star, \quad x_i \mid C \stackrel{\text{iid}}{\sim} \mathcal{N}(0, C), \quad C_e = \frac{1}{N} \sum_{i=1}^N x_i x_i^\top, \quad C \in \mathbb{S}_{++}^d. \quad (1)$$

Given $M \in \mathbb{N}$, we observe $C_{e,1}, \dots, C_{e,M}$ and know $N \in \mathbb{N}$, but we have no access to clean covariances nor pairs from the true joint law. Our aim is to estimate the prior $\pi_\star(dC)$ as well as the posterior $P_\star(dC \mid C_e)$ based on these observations.

Let $K_N(dC_e \mid C)$ denote the forward conditional law, also called the *Wishart channel*. Given any supplied C , we can sample this channel exactly: letting

$$C'_e = \frac{1}{N} C^{1/2} Z Z^\top C^{1/2}, \quad Z \in \mathbb{R}^{d \times N}, \quad Z_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (2)$$

we have $C'_e \sim K_N(\cdot \mid C)$. This sampler provides the extra information that we will use to learn $\pi_\star$ and $P_\star$. The other information comes from the observations $C_{e,1}, \dots, C_{e,M}$, which are samples from $\mu_\star(dC_e) \equiv \int K_N(dC_e \mid C) \pi_\star(dC)$.

If some covariate c can be associated with the observations C_e , we can straightforwardly to estimate $\pi_\star(dC \mid c_e)$ and $P_\star(dC \mid C_e, c)$. Appendix I gives this extension and its guarantees; below, c is constant.

Notation. We use C, C_e for random matrices and generic values when unambiguous. Subscript e denotes empirical quantities; hats denote fitted functions or estimates. $\text{diag}(v)$ forms a diagonal matrix from a vector v , and spec lists eigenvalues in increasing order. Write $\mathbb{S}^d$ for symmetric matrices and $\|A\|_F^2 = \text{tr}(A^T A)$.

2.2 AN IDEAL UPDATE WITH A MISSING CONDITIONAL SAMPLER

The posterior of any candidate prior π is

$$Q_\pi(dC | C_e) = \frac{k_N(C_e | C)\pi(dC)}{m_\pi(C_e)}, \quad m_\pi(C_e) = \int k_N(C_e | C)\pi(dC). \quad (3)$$

Here k_N is the density of the conditional law K_N relative to the common reference measure $\kappa_N = K_N(\cdot | I_d)$, i.e. such that $K_N(dC_e | C) = k_N(C_e | C) \kappa_N(dC_e)$ (see Appendix A for its expression).

Starting from an initial distribution π_0 , an ideal expectation-maximization iteration samples the conditional law $Q_k = Q_{\pi_k}$ associated with the current estimate π_k of the prior at iteration k (the E-step) and averages it over the observed population (the M-step) to obtain the next iterate π_{k+1} :

$$\pi_{k+1} = \mathcal{T}(\pi_k), \quad \mathcal{T}(\pi)(dC) = \int Q_\pi(dC | C_e) \mu_\star(dC_e). \quad (4)$$

This is *nonparametric EM*: the whole prior distribution is updated, rather than a fixed list of parameters (Aubin-Frankowski et al., 2022).

Equation (4) is not yet a practical algorithm since we must specify how to sample Q_k . We obtain that conditional sampler by training an SI on synthetic pairs drawn from π_k and the known forward mechanism. Applying it to the recorded observations supplies samples representing the next prior. Fresh re-corruption and conditional training then construct the next E-step. Section 3 gives a direct matrix implementation and its guarantees. Section 4 exploits RI to implement the same full-posterior objective with a smaller learned state.

3 GENERAL COVARIANCE RECONSTRUCTION

3.1 CONDITIONAL SAMPLING IN LOGARITHMIC COORDINATES

Given a covariance matrix C , we encode it as $Y = \Phi(C) = \text{svec}(\log C)$ where the map $\text{svec} : \mathbb{S}^d \rightarrow \mathbb{R}^p$, $p = d(d+1)/2$, lists the diagonal and $\sqrt{2}$ times the upper off-diagonal entries of a symmetric matrix, preserving its Frobenius norm. Thus Φ maps $\mathbb{S}_{++}^d$ bijectively onto $\mathbb{R}^p$. Its inverse is $C = \Phi^{-1}(Y) = \exp(\text{svec}^{-1} Y)$, positive definite for every finite Y . We take the logarithms before constructing the interpolation and exponentiate the generated endpoint. This changes the coordinates of inference, not its conditional target; regression and numerical errors can depend on that choice.

For $N \geq d$, use the same map for the observation: $Y_e = \Phi(C_e)$. For $N < d$, encode the singular observation as $Y_e = \Phi(C_e + \epsilon_0 I_d)$ with a fixed known $\epsilon_0 > 0$. The inverse recovers C_e by subtracting this shift, so the encoding retains all observation information and does not change the forward model. We use the same convention for Y'_e computed from a simulated observation.

Generate and record training pairs. For a current sampleable prior π , draw $C \sim \pi$ and a Gaussian matrix Z , calculate C'_e by (2), and record the encoded pair (Y'_e, Y) . Conditional on C , the Gaussian draw supplies the sampling noise; the empirical covariance and its encoding are then calculated. Pairs may be stored or generated as needed, with their values treated as fixed when differentiating the regression loss.

Fit the conditional drift. Throughout the implementation we choose $\alpha_t = 1 - t$, $\beta_t = t$, and $\sigma_t = \sigma(1 - t)$, with a positive scale σ . The conditional Brownian interpolant of Chen et al. (2024)

and its regression loss become

$$I_t = (1-t)Y'_e + tY + \sigma(1-t)W_t, \quad R_t = Y - Y'_e - \sigma W_t, \quad (5)$$

$$\mathcal{L}_\pi(\hat{b}) = \int_0^1 w(t) \mathbb{E}_{\pi_{K_N, W}} \|\hat{b}_t(I_t, Y'_e) - R_t\|_2^2 dt, \quad (6)$$

where W_t is standard Brownian motion in $\mathbb{R}^p$, independent of the pair, and $w(t) > 0$. We have $I_0 = Y'_e$ and $I_1 = Y$. The target R_t contains no derivative of Brownian motion. A training example needs only the Gaussian draw $W_t = \sqrt{t}z$, $z \sim \mathcal{N}(0, I_p)$, not a simulated path. The positive scale changes the interpolation and the fitted drift, not the exact posterior endpoint. Appendix B gives the derivation.

Sample and decode. At a supplied observation, retain Y_e as the conditioning input throughout the SDE. For the fitted drift $\hat{b}$, use the Föllmer noise coefficient $g_t^F = \sigma\sqrt{1-t^2}$ and the matching drift $\hat{b}_t^F(y, Y_e) = (1+t)\hat{b}_t(y, Y_e) - y + Y_e$:

$$\begin{aligned} dY_t &= [(1+t)\hat{b}_t(Y_t, Y_e) - Y_t + Y_e]dt + \sigma\sqrt{1-t^2}dB_t, \\ Y_0 &= Y_e, \quad C^{(r)} = \Phi^{-1}(Y_1). \end{aligned} \quad (7)$$

Each draw r uses fresh Brownian motion B_t . With an exact fitted drift, the simpler SDE with drift $\hat{b}_t^\pi$ and noise coefficient $\sigma(1-t)$ has the same conditional marginals. Fitted approximations need not agree. The displayed Föllmer choice minimizes the trajectory-KL error coefficient for a fixed drift approximation, under the assumptions of Chen et al. (2024); it is not a claim of optimal numerical accuracy. Changing the noise requires changing the drift as shown. Appendix B derives the coefficients and states endpoint hypotheses; Appendix B.7 describes numerical integration. The quadratic training loss fits a drift, not a covariance mean, and does not select a downstream estimation loss.

Choosing the noise scale. A starting choice is $\sigma = \kappa\sqrt{2/N}$ with a dimensionless tuning constant $\kappa > 0$. The $N^{-1/2}$ scale is motivated by scalar log-covariance sampling noise and by a fixed- d linearization near $C = cI_d$ (Appendix B.6); it is not a universal optimum. In spectral coordinates, dependence between empirical eigenvalues and the ratio d/N require separate calibration. We use a fixed scale for each run. Any fixed positive scale gives the same ideal EM map when the sampler assumptions hold, but finite regression and integration errors can depend strongly on it.

3.2 SELF-CONSISTENCY AND EXACT EM GUARANTEES

After reconstruction, decode each Y to $C = \Phi^{-1}(Y)$, draw fresh Gaussian noise, calculate C'_e by (2), and record (Y'_e, Y) for the next regression. Algorithm 1 repeats this construction. Reconstructed targets and simulated observations are fixed with respect to differentiation through the new network. Training again on the original observations paired with their reconstructions would simply reproduce the current conditional law in the exact-fitting limit. Fresh corruption instead gives pairs from the reconstructed prior and the known forward law (Modi et al., 2026).

Proposition 1 (The learned operation supplies the E-step). *Assume the regression moments, interior regularity, uniqueness, and endpoint limits in Appendix B. For synthetic pairs generated from π_{K_N} , the unrestricted regression minimizer is $b_t^\pi(y, Y_e) = \mathbb{E}_\pi[R_t \mid I_t = y, Y'_e = Y_e]$. With this minimizer in place of $\hat{b}$, the SDE (7) has conditional marginals equal to those of the interpolant and hence*

$$\Phi^{-1}(Y_1) \sim Q_\pi(\cdot \mid C_e). \quad (8)$$

The base SDE has the same endpoint under its corresponding hypotheses. Exact population reconstruction in Algorithm 1 therefore gives the EM update (4).

The proof conditions the interpolant’s drift in Itô’s formula and uses uniqueness of the density evolution. The full conditional law is produced by the SDE, not assumed available. “Population” means that expectations and conditional training are exact; it does not describe finite-data optimization.

Inputs: recorded $C_{e,1:M}$, N , a sampleable π_0 , a noise scale $\sigma > 0$, and an iteration budget K . Encode the recorded observations as $Y_{e,1:M}$.

1. **Initialize.** Draw $C \sim \pi_0$ and Gaussian noise, calculate C'_e using (2), and record $(Y'_e, \Phi(C))$. Fit $\hat{b}_0$ using (6).
2. **For** $k = 0, \dots, K-1$, **reconstruct.** Run (7) with $\hat{b}_k$ at sampled recorded observations. Decode each endpoint to a covariance C .
3. **Re-corrupt and refit.** Draw fresh Gaussian noise, calculate C'_e from each reconstructed C , and record $(Y'_e, \Phi(C))$. Fit $\hat{b}_{k+1}$ on these pairs, keeping their values fixed in network differentiation. Repeat steps 2–3 for the next k .

Full-matrix inference: at a supplied C_e , run the final conditional SDE with fresh noise and return $\Phi^{-1}(Y_1)$. Repeat for approximate posterior draws.

Algorithm 1: General covariance reconstruction by conditional SI. Every clean target is sampled from the current model, not supplied from the unknown true prior.

3.3 IDENTIFIABILITY: WHAT THE ENSEMBLE CAN DETERMINE

A prior is identifiable from the observations when no different prior produces the same distribution of empirical covariances. This is a property of the statistical problem, independent of the sampler or coordinate representation.

Theorem 2 (Sharp sample-size threshold). *For a known zero mean, $\pi \mapsto K_N \pi$ is injective on Borel probability measures on $\mathbb{S}_{++}^d$ if and only if $N \geq d$. When $N < d$, there are distinct smooth RI priors with common compact support inside $\mathbb{S}_{++}^d$ and identical observed laws.*

The Gaussian sample probes the prior’s Laplace transform only at matrices of rank at most N . For $N \geq d$ these include the positive-definite cone, which determines the prior; for $N < d$ the proof constructs perturbations invisible to these probes (Appendix C). The two indistinguishable priors also have different posterior kernels. Thus even infinitely many tasks cannot determine an unrestricted covariance posterior when $N < d$. This does not preclude recovery on a smaller identifiable class: known-basis diagonal priors are identifiable for every $N \geq 1$ (Proposition 7). Identifiability alone supplies neither a convergence rate nor a guarantee for a particular initialization.

3.4 CONVERGENCE OF THE IDEAL SI–EM ITERATION

For the remainder of the population analysis, assume Proposition 1 applies at every iteration, with exact regression, exact conditional SDE sampling, and the population observation average. The output laws are then exactly $\pi_{k+1} = \mathcal{T}(\pi_k)$, independently of $\sigma > 0$. The Kullback–Leibler divergence is $\text{KL}(P\|Q) = \int \log(dP/dQ) dP$ when $P \ll Q$, and $+\infty$ otherwise. Here $P \ll Q$ means that a set of probability zero under Q must also have probability zero under P . Let $D_k = \text{KL}(\pi_\star\|\pi_k)$ and $F_k = \text{KL}(\mu_\star\|K_N \pi_k)$.

Theorem 3 (Ideal EM progress and conditional reconstruction). *If $D_0 < \infty$, the exact population iteration satisfies*

$$\int \text{KL}(Q_\star(\cdot | C_e)\|Q_k(\cdot | C_e)) \mu_\star(dC_e) = D_k - F_k, \quad (9)$$

$$D_{k+1} \leq D_k - F_k, \quad F_k - F_{k+1} \geq \text{KL}(\pi_{k+1}\|\pi_k) \geq 0. \quad (10)$$

Consequently $\sum_{k \geq 0} F_k \leq D_0$ and $F_k \leq D_0/(k+1)$.

The proof uses the KL chain rule for the true and current joint laws, followed by posterior averaging (Appendix D). The $O(1/k)$ bound concerns the *observed* distribution. The identity for conditional reconstruction shows why this is not enough by itself: $F_k \rightarrow 0$ does not imply that $D_k - F_k$ tends to zero. The following conditions give prior and posterior recovery.

Corollary 4 (Prior and posterior recovery). *Assume Theorem 3. If $N \geq d$ and $\pi_0, \pi_\star$ are supported on a common compact set $\mathcal{K} \subset \mathbb{S}_{++}^d$, then*

$$\pi_k \Rightarrow \pi_\star, \quad Q_k(\cdot | C_e) \Rightarrow Q_\star(\cdot | C_e) \quad \text{for } \mu_\star\text{-almost every } C_e. \quad (11)$$

More generally the same conclusion holds on a common compact class on which K_N is injective. If, instead, an inverse-stability inequality $D_k \leq \chi F_k$ holds along the iterates with $\chi \geq 1$, then

$$D_k \leq (1 - \chi^{-1})^k D_0, \quad \int \text{KL}(Q_\star \| Q_k) d\mu_\star = D_k - F_k \leq (1 - \chi^{-1})^k D_0. \quad (12)$$

Here $\Rightarrow$ means convergence of expectations of bounded continuous functions. Compact support can, for example, impose fixed positive lower and upper eigenvalue bounds. The posterior conclusion follows because the likelihood is positive and continuous in C on this compact set; its proof is included in Appendix D. It is weak convergence, not convergence of prior or posterior KL. The stronger inverse-stability premise controls how much discrepancy can be hidden by the observation channel; it is a separate assumption, not a consequence of $N \geq d$, and χ depends on the permitted priors as well as N, d .

Initialization matters. Exact EM preserves absolute continuity: $\pi_{k+1} \ll \pi_k$. A finite set of initial particles therefore cannot acquire new support under exact posterior averaging. Every point-mass prior is an EM fixed point, even when incorrect. The assumption $D_0 < \infty$ excludes these incorrect fixed points but is stronger than merely assigning positive mass to every open set. It also rules out a purely continuous initialization for an atomic truth. These qualifications are part of the ideal convergence statements, not effects of imperfect drift learning.

Local convergence and the role of N . For a bounded centered perturbation of $\pi_\star$, let $A_N h(C_e) = \mathbb{E}[h(C) \mid C_e]$. The derivative of the EM map is $\text{Id} - A_N^* A_N$, where A_N^* denotes its adjoint. A direction with $A_N^* A_N h = \eta h$ contracts locally by $1 - \eta$: features poorly visible in C_e converge slowly. Appendix E defines the function spaces, proves this statement and monotonicity of information with N , and gives an explicit inverse-gamma example. These characterize ideal EM, not neural optimization.

Beyond exact learning. The main results assume that each conditional drift is learned exactly. Appendix G retains a conditional one-update bound for an incompletely fitted drift, including its dependence on σ , and a finite-iteration comparison with ideal EM. These require weighted population regression error and an observation-overlap bound. They neither prove that a network attains those errors nor establish recovery from a fixed finite dataset. We therefore treat them as supplementary results; practical accuracy must be measured separately.

4 RECONSTRUCTION UNDER ROTATIONAL INVARIANCE

4.1 ONLY THE SPECTRAL POSTERIOR MUST BE LEARNED

A prior is RI if C and $Q C Q^T$ have the same law for every orthogonal Q . Write

$$C = U \Lambda U^T, \quad C_e = U_e \Lambda_e U_e^T, \quad R = U_e^T U. \quad (13)$$

Here $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ and $\Lambda_e = \text{diag}(\lambda_{e,1}, \dots, \lambda_{e,d})$ list eigenvalues in increasing order. We regard a spectral law ν as a law on these ordered diagonal matrices. It determines the entire RI prior. It can be generated by drawing $\Lambda \sim \nu$ and U uniformly over orthogonal matrices, independently. The posterior has the representation

$$Q_\nu(d\Lambda, dR \mid \Lambda_e) \propto \nu(d\Lambda) |\Lambda|^{-N/2} e^{-\mathcal{V}_{\Lambda, \Lambda_e}(R)} dH(R), \quad (14)$$

$$\mathcal{V}_{\Lambda, \Lambda_e}(R) = \frac{N}{2} \text{tr}(\Lambda^{-1} R^T \Lambda_e R).$$

where H is the uniform, or Haar, probability measure on orthogonal matrices. In particular, the conditional orientation law

$$H_N(dR \mid \Lambda, \Lambda_e) \propto e^{-\mathcal{V}_{\Lambda, \Lambda_e}(R)} dH(R) \quad (15)$$

does not depend on ν . Only $q_\nu(d\Lambda \mid \Lambda_e)$, the spectral posterior, must be learned. Write $K_N^{\text{spec}}(\cdot \mid \Lambda)$ for the forward law of Λ_e given Λ ; its simulator is specified below. Under RI, the unknown prior is determined by the joint law of its ordered eigenvalues. Exact spectral EM is therefore full-matrix EM restricted to RI priors; prior and observation KL discrepancies are unchanged. Appendix F, Theorem 9, gives the precise reduction and proof. This does not remove the $N \geq d$ threshold for unrestricted RI priors, nor make the conditional orientation uniform. We now use the reduction to implement and analyze full covariance reconstruction.

4.2 SPECTRAL SI AND THE SECOND LEARNING ALGORITHM

Use the clean vector $Y = (\log \lambda_i)_{i=1}^d$ and the observation vector $Y_e = (\log \lambda_{e,i})_{i=1}^d$. When the observation has zero eigenvalues, use $Y_e = (\log(\lambda_{e,i} + \epsilon_0))_{i=1}^d$. The drift acts jointly on all coordinates, conditioned on the entire empirical spectrum; diagonal storage does not assume independent eigenvalues. For a supplied Λ , draw Gaussian Z and calculate

$$\Lambda'_e = \text{diag} \left[\text{spec} \left(\frac{1}{N} \Lambda^{1/2} Z Z^T \Lambda^{1/2} \right) \right], \quad K_N^{\text{spec}}(\cdot | \Lambda) = \text{Law}(\Lambda'_e | \Lambda). \quad (16)$$

Record (Y'_e, Y) and fit (6) using the same schedules as Section 3:

$$I_t = (1 - t)Y'_e + tY + \sigma(1 - t)W_t, \quad R_t = Y - Y'_e - \sigma W_t, \quad W_t \in \mathbb{R}^d. \quad (17)$$

Run the Föllmer sampling rule (7) in dimension d . Exponentiate the endpoint coordinates, sort them in increasing order, and place them on the diagonal of Λ . Sorting only decodes the endpoint; the interior SDE is unchanged. Exact endpoints already have the chosen order. Log-gap coordinates enforce strict order as an alternative, but cannot represent repetitions (Appendix B.1). SI regularity and finite-error assumptions must be checked separately for lower-dimensional spectral priors.

Inputs: empirical spectra $\Lambda_{e,1:M}$, N , a sampleable spectral prior ν_0 , a noise scale $\sigma > 0$, and a budget K .

1. **Initialize.** Draw $\Lambda \sim \nu_0$ and Gaussian noise, calculate Λ'_e by (16), and record (Y'_e, Y) . Fit the drift using (6).
2. **For** $k = 0, \dots, K - 1$: run the spectral SI at recorded $\Lambda_{e,a}$ with independent noise, decode the population spectra, and simulate fresh Λ'_e from them. Refit the next drift using (17), with reconstructed targets held fixed.

Full-matrix inference at $C_e = U_e \Lambda_e U_e^T$:

1. Draw Λ with the final spectral SI conditioned on Λ_e .
2. With this Λ fixed, run orientation MCMC targeting (15); return $C = U_e \Lambda \Lambda^T U_e^T$.

No orientation sampling occurs during spectral training.

Algorithm 2: RI reconstruction: self-consistent spectral learning followed by a known conditional orientation sampler. Both algorithms target the full covariance posterior.

The posterior-regression proof from Proposition 1 applies with K_N^{spec} . Thus exact spectral learning implements

$$\nu_{k+1}(d\Lambda) = \int q_{\nu_k}(d\Lambda | \Lambda_e) \mu_{\star}^{\text{spec}}(d\Lambda_e), \quad \mu_{\star}^{\text{spec}} = K_N^{\text{spec}} \nu_{\star}. \quad (18)$$

Theorem 3 applies to $(\nu, \Lambda, \Lambda_e)$ under exact population learning. The full-posterior recovery consequences are stated below. Simulation still requires a sample-matrix eigendecomposition; it is not the independent chi-square channel of a covariance diagonal in a known basis. Only the learned state, not every computational cost, has been reduced to dimension d .

4.3 COMPLETING THE MATRIX BY A KNOWN CONDITIONAL SAMPLER

The density (15) is a matrix Bingham density: its logarithm is a quadratic expression in orthogonal-matrix entries. Conditional-update methods are available (Hoff, 2009). Markov chain Monte Carlo (MCMC) uses random updates whose limiting law is this target. No SI or prior-density evaluation is needed. A concrete Metropolis kernel proposes $R' = R G_{ij}(\theta)$, where G_{ij} rotates two columns and θ has a symmetric proposal law, and accepts with probability

$$\min\{1, \exp[\mathcal{V}_{\Lambda, \Lambda_e}(R) - \mathcal{V}_{\Lambda, \Lambda_e}(R')]\}. \quad (19)$$

The unknown normalizing constant cancels. For an explicit convergence guarantee, we also allow an independent Haar proposal with probability $\omega > 0$, using the same acceptance rule. Appendix F.4 proves invariance and convergence of this mixture. The global proposal provides a simple guarantee, not a claim of fast mixing; local or block updates can be more efficient in practice.

After m steps, denote the conditional output law by $\hat{H}_m(\cdot \mid \Lambda, \Lambda_e)$. The returned sample is

$$\Lambda \sim \hat{q}_{\text{num}}(\cdot \mid \Lambda_e), \quad R \sim \hat{H}_m(\cdot \mid \Lambda, \Lambda_e), \quad C = U_e R \Lambda R^T U_e^T. \quad (20)$$

Keep Λ fixed during each conditional orientation run. In general $U_e \Lambda U_e^T$ is not a matrix posterior draw. A factored output (U_e, R, Λ) represents the full matrix without forming a dense matrix product. Re-corruption and occasional global proposals may still cost cubic time with dense linear algebra. A finite MCMC run has a separate approximation error.

4.4 IDEAL SPECTRAL EM AND FULL-POSTERIOR RECOVERY

Theorem 5 (Exact completion preserves posterior error). *Fix $C_e = U_e \Lambda_e U_e^T$. Complete a spectral law $\hat{q}(\cdot \mid \Lambda_e)$ with the exact H_N , and call its matrix law $\hat{Q}^H$. Then*

$$\boxed{\text{KL}(Q_\star(\cdot \mid C_e) \parallel \hat{Q}^H(\cdot \mid C_e)) = \text{KL}(q_\star(\cdot \mid \Lambda_e) \parallel \hat{q}(\cdot \mid \Lambda_e))}. \quad (21)$$

The identity includes repeated eigenvalues when the reconstructed matrix law is used, without requiring a unique eigenbasis.

The spectrum is recoverable from a reconstructed matrix, and the exact conditional matrix law given it is common to both models. The KL chain rule therefore gives (21), with no dimension-dependent loss (Appendix F.2). Finite MCMC introduces a separate orientation error, bounded in Proposition 10.

Let ν_k be the ideal spectral EM iterates, $D_k^\nu = \text{KL}(\nu_\star \parallel \nu_k)$, and $F_k^\nu = \text{KL}(\mu_\star^{\text{spec}} \parallel K_N^{\text{spec}} \nu_k)$. Write Q_k^H for the full matrix law obtained by completing q_{ν_k} with exact H_N .

Corollary 6 (Convergence of the ideal RI reconstruction). *Assume exact spectral regression and exact SDE sampling at each population iteration, with the sampler hypotheses of Proposition 1, and $D_0^\nu < \infty$. Then*

$$F_k^\nu \leq \frac{D_0^\nu}{k+1}, \quad \int \text{KL}(Q_\star \parallel Q_k^H) d\mu_\star = D_k^\nu - F_k^\nu. \quad (22)$$

If $N \geq d$ and $\nu_0, \nu_\star$ have support in a common compact set of positive spectra, then $\nu_k \Rightarrow \nu_\star$ and $Q_k^H(\cdot \mid C_e) \Rightarrow Q_\star(\cdot \mid C_e)$ for $\mu_\star$ -almost every observation. If $D_k^\nu \leq \chi F_k^\nu$ instead holds along the iterates, then the averaged full-posterior KL in (22) is at most $(1 - \chi^{-1})^k D_0^\nu$.

The proof lifts the spectral iterates to RI matrix laws and applies Corollary 4; exact completion is their full posterior (Appendix F.3). Thus the reduced algorithm has the same ideal recovery objective and guarantees as the general one. Finite orientation sampling is treated separately in Appendix F.4, without a useful dimension-uniform rate. Incomplete spectral learning and time discretization are covered only by the conditional supplementary bound in Appendix G.8.

4.5 COVARIANCE ESTIMATION AND THE LW/LP COMPARISON

Both methods learn distributions before choosing a decision loss. Under squared Frobenius loss the posterior mean is optimal; for Stein loss $L_S(A, C) = \text{tr}(C^{-1}A) - \log \det(C^{-1}A) - d$, the optimal estimate instead averages precision and then inverts:

$$\hat{C}_F = \mathbb{E}[C \mid C_e], \quad \hat{C}_S = (\mathbb{E}[C^{-1} \mid C_e])^{-1}. \quad (23)$$

Under RI these share the empirical eigenvectors. In particular,

$$\hat{C}_F = U_e \text{diag}(m(\Lambda_e)) U_e^T, \quad m_i(\Lambda_e) = \mathbb{E}[\xi_i \mid \Lambda_e], \quad \xi_i = u_{e,i}^T C u_{e,i} = \sum_j \lambda_j R_{ij}^2. \quad (24)$$

Thus $\mathbb{E}[\lambda_i \mid \Lambda_e]$ estimates a population eigenvalue, whereas $m_i(\Lambda_e)$ estimates variance along an empirical direction. LW/LP covariance shrinkage approximates the latter directional oracle ξ_i , not generally the mean population spectrum (Ledoit & Péché, 2011; Ledoit & Wolf, 2012). Appendix F.6 gives moment assumptions, Stein coefficients, and alternatives for computing estimates. Proposition 12 states a precise averaged connection to LP-type coefficients. Neither shrinkage coefficients nor posterior means replace population-spectrum draws in Algorithm 2. The quadratic SI loss learns a sampling drift; it does not choose a covariance-estimation loss.

5 PROPOSED NUMERICAL EXPERIMENTS

This section specifies evaluation protocols, not completed results. The main comparison is between Algorithms 1 and 2 on the same RI posterior problem; “general” does not mean “non-RI.” The general implementation also serves when the prior has recurring directional structure. We propose the following tests, with common task-level train/test splits and reported seed variability.

Controlled validation. Use scalar and known-basis diagonal examples with accurately resolved posterior integrals to check the ideal EM iteration, its initialization dependence, and the inverse-gamma mode in Proposition 8. Then use supervised synthetic pairs to isolate conditional regression and SDE integration before hiding the clean targets and testing the full loop. Compare base and Föllmer samplers at equal evaluation budgets and vary M , N , training effort, and outer iterations separately. Sweep positive noise scales, including $\sigma = \kappa\sqrt{2/N}$, without presuming this baseline optimal.

Full posterior reconstruction under RI. At feasible dimensions, compare both algorithms on identical RI priors and observed ensembles. Validate covariance entries and directional variances $v^T C v$, not only trace, log determinant, or eigenvalues: spectral summaries alone cannot validate orientation. Appendix H specifies inverse-Wishart and mixture benchmarks with explicit full posteriors, reference sampling, and conditional accuracy tests. Include identifiable cases and the indistinguishable $d = 2$, $N = 1$ priors in Appendix C.3; failure to distinguish the latter is an observation limit, not a training failure.

Orientation error and computational scaling. Hold spectral draws fixed and vary the MCMC run length independently of the SDE grid. Check multiple starts and effective sample sizes, which account for correlation. Report network and simulator calls, eigendecompositions, orientation time, and total time per effectively independent full covariance draw. Separate preprocessing, training, and inference; do not omit orientation cost. Compare the guaranteed but potentially slow Haar-mixture sampler with other updates without claiming an unproved rate for them.

Non-RI priors and covariance estimation. Use a population with shared directions or structured factors to test full-matrix recovery. The RI model is then a restricted comparison. Report posterior interval coverage and predictive checks separately from Frobenius or Stein risk. Compare with sample covariance, linear LW, nonlinear shrinkage, an ensemble-aware covariance estimator, and conditional diffusion/EM. True-prior and supervised samplers use additional information and are reference benchmarks. Covariance/PCA comparisons use $m_i(\Lambda_e)$, not mean population eigenvalues. Appendix I adds conditional benchmarks and measured-data proposals. Architectures, budgets, and measured results remain to be supplied.

6 DISCUSSION

The common problem is posterior sampling without the true prior or clean training pairs. Section 2 states the ideal update; the two algorithms supply its missing conditional sampler. The general method learns all matrix-log coordinates. RI permits learning spectra and completing full matrices using an explicit conditional orientation law, with separate spectral and MCMC errors. Neither approach assumes the posterior available, and neither is prior-free: the ensemble is used to learn the unknown prior. The principal guarantees concern identifiability and the ideal population iteration with exactly learned drifts: observation convergence is unconditional within the finite- D_0 premise, while prior and posterior recovery require the stated compactness or inverse-stability conditions. Supplementary bounds describe incomplete learning but do not prove neural-training convergence. Non-identifiability, initialization, finite data, endpoint error, and slow orientation mixing remain substantive limits. The proposed experiments must establish practicality; the theory alone does not do so.

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A THE GAUSSIAN EXPERIMENT AND ITS COVARIANCE CHANNEL

A positive-semidefinite (PSD) matrix can have zero eigenvalues; a positive-definite matrix has strictly positive eigenvalues. The word *channel* denotes the known conditional distribution of an observation given a supplied clean covariance. To write one likelihood valid even when $N < d$, use the probability measure $\kappa_N = K_N(\cdot | I_d)$ as reference. Every covariance-sampling law has a density relative to it:

$$k_N(C_e | C) = \frac{dK_N(\cdot | C)}{d\kappa_N}(C_e) = |C|^{-N/2} \exp\left[-\frac{N}{2} \text{tr}((C^{-1} - I_d)C_e)\right]. \quad (25)$$

Thus $K_N\pi = m_\pi\kappa_N$ with m_π defined in (3). This reference measure avoids treating a singular empirical covariance as if it had a density on the full positive-definite cone.

Let $X = (x_1, \dots, x_N) \in \mathbb{R}^{d \times N}$. Relative to the Gaussian experiment with covariance I_d , its likelihood ratio is

$$\frac{p_C(X)}{p_{I_d}(X)} = |C|^{-N/2} \exp\left[-\frac{1}{2} \text{tr}((C^{-1} - I_d)XX^T)\right] = k_N(C_e | C). \quad (26)$$

This is a positive function of $C_e = XX^T/N$ alone. Under the reference experiment, take the conditional law of X given C_e , written $L_N(dX | C_e)$. Equation (26) shows that the same conditional kernel applies under every C , and hence every mixing prior. Thus

$$\text{Law}_\pi(X) = \int L_N(\cdot | C_e)(K_N\pi)(dC_e). \quad (27)$$

In particular, the raw-sample distribution and the empirical-covariance distribution determine one another in this family. This argument does not require $N \geq d$, nor a nonsingular observed covariance. It proves the expression in (25).

For $N \geq d$, C_e is positive definite almost surely and its Lebesgue density on $\mathbb{S}_{++}^d$ is

$$\frac{dK_N(\cdot | C)}{dC_e}(C_e) = \frac{N^{Nd/2}}{2^{Nd/2}\Gamma_d(N/2)} |C_e|^{(N-d-1)/2} |C|^{-N/2} \exp\left[-\frac{N}{2} \text{tr}(C^{-1}C_e)\right], \quad (28)$$

where $\Gamma_d(a) = \pi^{d(d-1)/4} \prod_{j=1}^d \Gamma(a + (1-j)/2)$. For $N < d$ this Lebesgue density is unavailable, but (25) remains valid. All posterior formulas in the paper may be read relative to κ_N .

Noise moments. Gaussian fourth moments yield

$$\mathbb{E}[C_e | C] = C, \quad \text{Cov}((C_e)_{ij}, (C_e)_{kl} | C) = \frac{1}{N}(C_{ik}C_{jl} + C_{il}C_{jk}). \quad (29)$$

For $E_C = C^{-1/2}(C_e - C)C^{-1/2}$, the conditional law does not depend on C and

$$\mathbb{E}[\|E_C\|_F^2 | C] = \frac{d(d+1)}{N}. \quad (30)$$

The diagonal entries contribute $2d/N$ and the off-diagonal entries contribute $d(d-1)/N$. Differentiating the log likelihood in a symmetric direction H gives score $\frac{N}{2} \text{tr}(C^{-1}HC^{-1}(C_e - C))$ and Fisher information

$$\mathcal{I}_C(H, H) = \frac{N}{2} \text{tr}(C^{-1}HC^{-1}H). \quad (31)$$

These finite-dimensional noise and information formulas do not imply a uniform inverse-stability bound for a distribution of covariances. The latter is an infinite-dimensional inverse problem.

Centering and degrees of freedom. All stated thresholds use a known zero mean and the normalization in (1). If a Gaussian sample mean is subtracted and only the centered covariance is retained, the Wishart degrees of freedom become $N - 1$. After the corresponding known rescaling, the threshold uses those degrees of freedom rather than the raw number of observations. This does not cover a model with an additional unknown distribution of task means.

B SUPERVISED CONDITIONAL SAMPLING AND THE FÖLLMER PROCESS

This appendix supplies the proofs and general-schedule details for the conditional sampler specified in Section 3. Both main algorithms repeatedly use this supervised construction on synthetic pairs, in their respective dimensions. The Brownian-interpolant construction and its tunable diffusion follow Chen et al. (2024); we give the algebra in covariance-compatible coordinates and state the qualifications needed here. The distribution-level EM results do not require this particular sampler.

B.1 COORDINATES, DECODING, AND SCOPE

Sections 3 and 4 use Y_e for the retained observation expressed in numerical coordinates. We use that same conditioning argument throughout this appendix. Since the input encoding is injective, conditioning on Y_e is equivalent to conditioning on the retained observation. In the spectral and known-basis diagonal versions, that retained observation is Λ_e and c_e , respectively. A different deterministic initialization, such as zero, is also possible provided the observation is supplied separately to the drift.

For full matrices, svec is the linear isometry from $\mathbb{S}^d$ onto $\mathbb{R}^p$ obtained from the Frobenius-orthonormal symmetric basis. In particular $\|\text{svec}(A)\|_2 = \|A\|_F$ for every symmetric A , whether or not A is positive definite. We set $Y = \Phi(C) = \text{svec}(\log C)$ and $C = \Phi^{-1}(Y) = \exp(\text{svec}^{-1} Y)$. This is a bijection from $\mathbb{S}_{++}^d$ to $\mathbb{R}^{d(d+1)/2}$. If $\rho_j = \text{Law}(\Phi(C))$ for $C \sim \pi_j$, then $d\rho_1/d\rho_2(y) = d\pi_1/d\pi_2(\Phi^{-1}(y))$ whenever these derivatives exist. Integrating their logarithms gives equality of KL divergences; applying the inverse map also shows that posterior averaging commutes with the coordinate transformation. When absolute continuity fails, the corresponding KL values on both sides are infinite. This proves the coordinate-invariance assertion in Section 3.1. The same argument applies to the componentwise logarithm in the known-basis diagonal model.

For $N \geq d$, one can take $Y_e = \text{svec}(\log C_e)$. If $N < d$, $Y_e = \text{svec}(\log(C_e + \epsilon_0 I_d))$ with fixed $\epsilon_0 > 0$ is an injective encoding: $C_e = \exp(\text{svec}^{-1} Y_e) - \epsilon_0 I_d$. It changes neither the Wishart likelihood nor the information in C_e . In particular, the shift cannot resolve the non-identifiability of Theorem 2.

Frobenius Brownian motion on symmetric matrices has independent standard Brownian coefficients in the basis E_{ii} and $(E_{ij} + E_{ji})/\sqrt{2}$, $i < j$. Its off-diagonal entries have variance $t/2$, not t . This is equivalent to standard Brownian motion in the svec coordinates. Additive noise does not preserve positive definiteness in the original matrix space. In log coordinates, every finite decoded numerical output is positive definite and can be passed through the next forward simulation. This is a numerical advantage; exact posterior inference is unchanged by the bijection.

For RI, the primary target encoding in Section 4 is $Y_i = \log \lambda_i$ with increasing eigenvalue order. It is one-to-one on the ordered spectral state space, but that space is a subset of $\mathbb{R}^d$. The exact endpoint stays in it; a numerical endpoint may be unordered. Decoding by sorting $(e^{Y_i})_i$ always gives a positive ordered spectrum, and this same order is used to form the next training target. Sorting is applied after sampling, not by modifying the interior SDE. It has no effect on the exact endpoint. For an approximate endpoint, any deterministic decoding can only decrease its KL discrepancy from the exact decoded law by data processing. The spectral version of the one-update bound therefore remains an inequality; it need not use equality of KL under the extended sorting map.

An alternative for *strictly* ordered spectra uses log-positive-gap coordinates:

$$Y_1 = \log \lambda_1, \quad Y_j = \log(\lambda_j - \lambda_{j-1}) \quad (j \geq 2), \quad \lambda_j = \sum_{i=1}^j e^{Y_i}. \quad (32)$$

This gap parametrization is a bijection onto $\mathbb{R}^d$ and enforces strict ordering for every output. It cannot represent repeated eigenvalues exactly. Plain ordered log eigenvalues can represent repetitions, but their law may then have lower-dimensional support. Neither representation alone supplies the smooth-density and endpoint-relative-entropy hypotheses below. For observed spectra with zeros, the injective input $(\log(\lambda_{e,i} + \epsilon_0))_{i=1}^d$ remains valid regardless of which clean spectral coordinates are chosen. The drift is joint across coordinates in both versions.

For the following identities, fix Y_e and write $q(\cdot \mid Y_e)$ for the conditional density of Y . Assume the stated conditional expectations exist, the interior marginal densities and drifts are sufficiently regular for the Fokker–Planck calculation, and the relevant SDE and Fokker–Planck solutions are

unique in the considered class. Endpoint assertions require convergence to the intended endpoint law; finite trajectory-KL assertions additionally require the Girsanov absolute-continuity and integrability conditions. Girsanov’s formula expresses the likelihood ratio between SDE trajectory distributions with the same noise coefficient and different drifts. These are assumptions on the sampler, not consequences of arbitrary π_* . Atomic or lower-dimensional target priors can satisfy the EM theory while falling outside these density and finite-entropy conditions.

B.2 BROWNIAN INTERPOLANT AND DRIFT REGRESSION

With W_t standard Brownian motion in $\mathbb{R}^p$, independent of (Y_e, Y) , define

$$I_t = \alpha_t Y_e + \beta_t Y + \sigma_t W_t, \quad R_t = \dot{\alpha}_t Y_e + \dot{\beta}_t Y + \dot{\sigma}_t W_t, \quad (33)$$

where $\alpha_0 = \beta_1 = 1$ and $\alpha_1 = \beta_0 = \sigma_1 = 0$. Take $\beta_t, \sigma_t > 0$ for $0 < t < 1$ and schedules differentiable there with suitable endpoint limits. Since $W_0 = 0$, σ_0 need not vanish. The minimizer of the pointwise conditional square loss is

$$b_t(y, Y_e) = \mathbb{E}[R_t \mid I_t = y, Y_e]. \quad (34)$$

Equivalently, it minimizes $\int_0^1 w(t) \mathbb{E} \|\widehat{b}_t(I_t, Y_e) - R_t\|_2^2 dt$ whenever this loss is finite and $w(t) > 0$ on the interior. Uniform weighting is possible for regular schedules with finite integrated second moments.

Itô’s formula gives $dI_t = R_t dt + \sigma_t dW_t$. Thus, for a smooth compactly supported test function f ,

$$\frac{d}{dt} \mathbb{E}[f(I_t) \mid Y_e] = \mathbb{E}[\nabla f(I_t)^T b_t(I_t, Y_e) + \frac{1}{2} \sigma_t^2 \Delta f(I_t) \mid Y_e]. \quad (35)$$

Writing $\rho_t(\cdot \mid Y_e) = \text{Law}(I_t \mid Y_e)$, the conditional density solves

$$\partial_t \rho_t = -\nabla \cdot (b_t \rho_t) + \frac{1}{2} \sigma_t^2 \Delta \rho_t. \quad (36)$$

Uniqueness then identifies these marginals with those of $dY_t = b_t(Y_t, Y_e)dt + \sigma_t dB_t$, $Y_0 = Y_e$, where B_t is a fresh Brownian motion. Its endpoint is $q(\cdot \mid Y_e)$ under the stated endpoint assumptions. No derivative of Brownian motion appears in the regression target. This proves the base-SDE part of Proposition 1 after applying the decoding map (and Φ^{-1} in the full-matrix case). The drift correction below proves the displayed Föllmer version. The same projection argument holds for synthetic spectral or diagonal pairs because it only uses their joint law, not a matrix-specific likelihood. Because $k_N(C_e \mid C) > 0$ for every SPD C and κ_N -almost every observation, every probability mixture $K_N \pi$ is equivalent to κ_N . Thus the conditional identities established for synthetic observations hold μ_* -almost everywhere as well. Equivalence alone does not bound their approximation error after this change of observation distribution.

For any admissible fitted drift, conditional orthogonality also gives

$$\mathbb{E}[\|\widehat{b}_t(I_t, Y_e') - R_t\|_2^2] = \mathbb{E}[\|b_t^\pi(I_t, Y_e') - R_t\|_2^2] + e_t(\pi, \widehat{b}). \quad (37)$$

Integrating with a positive time weight proves the excess-risk identity used in Appendix G. The weights are deterministic, so whenever the objectives are finite they do not change the unrestricted pointwise minimizer. A finite architecture or finite optimization budget can of course react differently to different weights.

B.3 TUNABLE DIFFUSION AND RECOVERY OF THE SCORE

The volatility is the coefficient multiplying Brownian noise. For a suitable deterministic scalar volatility g_t , set

$$b_t^g(y, Y_e) = b_t(y, Y_e) + \frac{1}{2}(g_t^2 - \sigma_t^2) \nabla_y \log \rho_t(y \mid Y_e). \quad (38)$$

Substitution into the Fokker–Planck equation shows that $dY_t^g = b_t^g(Y_t^g, Y_e)dt + g_t dB_t$ has the same marginals whenever this equation and its initial condition are well posed. The drift correction is essential. In particular, setting $g = 0$ does not provide a regular deterministic ODE that spreads a point mass; singular behavior would have to be treated separately.

The score is the gradient of the log conditional density, $\nabla_y \log \rho_t$. It can be recovered from b_t when the following denominator is nonzero:

$$\zeta_t = [t\sigma_t(\dot{\beta}_t\sigma_t - \beta_t\dot{\sigma}_t)]^{-1}, \quad j_t(y, Y_e) = \dot{\beta}_ty + (\beta_t\dot{\alpha}_t - \dot{\beta}_t\alpha_t)Y_e. \quad (39)$$

Indeed, Gaussian differentiation and (33) give

$$\nabla_y \log \rho_t(y | Y_e) = -\frac{\mathbb{E}[W_t | I_t = y, Y_e]}{t\sigma_t}, \quad (40)$$

$$\beta_t b_t(y, Y_e) - j_t(y, Y_e) = (\beta_t\dot{\sigma}_t - \dot{\beta}_t\sigma_t)\mathbb{E}[W_t | I_t = y, Y_e], \quad (41)$$

so

$$\nabla_y \log \rho_t(y | Y_e) = \zeta_t[\beta_t b_t(y, Y_e) - j_t(y, Y_e)]. \quad (42)$$

This algebra permits changing g after learning b . It does not remove the need to handle apparent endpoint singularities in ζ_t . The endpoint regularity conditions in Chen et al. (2024), or directly verified conditions for a chosen schedule, are needed in applying these identities on the closed time interval.

B.4 PATH-SPACE ERROR AND THE GENERAL FÖLLMER CHOICE

Define the approximate $\hat{b}^g$ by substituting $\hat{b}$ into both (38) and (42). Then

$$\hat{b}_t^g - b_t^g = [1 + \frac{1}{2}\beta_t\zeta_t(g_t^2 - \sigma_t^2)](\hat{b}_t - b_t). \quad (43)$$

Let P^g and $\hat{P}^g$ denote the conditional path laws, meaning probability distributions over entire trajectories, of the exact and approximate SDEs with the same initial point and volatility. Under Girsanov's hypotheses,

$$\text{KL}(P^g \parallel \hat{P}^g) = \frac{1}{2} \int_0^1 \frac{[1 + \frac{1}{2}\beta_t\zeta_t(g_t^2 - \sigma_t^2)]^2}{g_t^2} L_t dt, \quad L_t = \mathbb{E}[\|\hat{b}_t(I_t, Y_e) - b_t(I_t, Y_e)\|_2^2 | Y_e]. \quad (44)$$

The same identity holds on a truncated interval under the corresponding restricted hypotheses. Marginalization bounds endpoint KL by this path KL. Since L_t does not depend on g_t , minimize the prefactor pointwise: writing $a = \beta_t\zeta_t/2$ and $c = 1 - a\sigma_t^2$, the relevant function is $(c + ag_t^2)^2/g_t^2$. For nondegenerate a, c , its minimum occurs at $g_t^2 = |c/a|$. Hence

$$(g_t^F)^2 = \left| 2t\sigma_t \left(\frac{\dot{\beta}_t}{\beta_t}\sigma_t - \dot{\sigma}_t \right) - \sigma_t^2 \right|. \quad (45)$$

Zero or singular minimizing values need a limiting or separate well-posedness argument. When $\beta_t/(\sqrt{t}\sigma_t)$ is increasing, the quantity inside the absolute value is nonnegative. For the implemented choice $\alpha_t = 1 - t$, $\beta_t = t$, and $\sigma_t = \sigma(1 - t)$, this gives $(g_t^F)^2 = \sigma^2(1 - t^2)$.

The Föllmer process minimizes relative entropy of the trajectory distribution, subject to its prescribed endpoint law and relative to a given reference process. For general schedules that reference is *specified*, not automatically standard Brownian motion. With $v_t = \beta_t^2 + t\sigma_t^2$, the reference in Chen et al. (2024) is

$$dZ_t = [\dot{\alpha}_t Y_e + a_t(Z_t - \alpha_t Y_e)]dt + g_t^F dB_t, \quad a_t = \frac{d}{dt} \log \frac{v_t}{\beta_t}, \quad Z_0 = Y_e. \quad (46)$$

Under the required regularity, monotonicity, and finite terminal relative entropy, the exact process using (45) is the path-relative-entropy minimizer with terminal density $q(\cdot | Y_e)$ for this reference. This is distinct from optimizing the outer EM iteration. It does not establish that g^F is optimal for numerical discretization, covariance risk, or finite-data posterior calibration.

B.5 CLOSED-FORM FÖLLMER DRIFT FOR THE MAIN IMPLEMENTATION

For the positive scale retained throughout the main implementation, $\alpha_t = 1 - t$, $\beta_t = t$, and $\sigma_t = \sigma(1 - t)$ with $\sigma > 0$, the score-recovery coefficients reduce to $\zeta_t = [\sigma^2 t(1 - t)]^{-1}$ and $j_t(y, Y_e) = y - Y_e$. Formula (45) gives $(g_t^F)^2 = \sigma^2(1 - t^2)$, so

$$\frac{1}{2}((g_t^F)^2 - \sigma_t^2) = \sigma^2 t(1 - t). \quad (47)$$

Substituting the recovered score into (38) therefore yields

$$b_t^{\pi, F} = b_t^\pi + \sigma^2 t(1-t)\zeta_t[tb_t^\pi - y + Y_e] = (1+t)b_t^\pi - y + Y_e. \quad (48)$$

The same algebra gives the fitted drift in (7) for every $\sigma > 0$. The scale cancels from this explicit drift correction, but changes the training distribution and hence the learned function itself. Unlike the general score formula, this expression has no explicit division by t or $1-t$, and requires one evaluation of the fitted drift per integration step. The process remains subject to the regularity and integrability hypotheses stated above. Comparing its time grid and error with the base sampler is an empirical and numerical-analysis question, not resolved by the continuous-time path-KL optimum.

B.6 CHOOSING THE SCALE OF THE INTERPOLATION NOISE

The scale σ is an algorithmic parameter in the chosen coordinates, not an additional variance in the observation model. For every fixed $\sigma > 0$ satisfying the sampler assumptions, exact conditional training gives the same posterior endpoint and hence the same population EM map. A dependence on N is nevertheless natural for the numerical problem.

In the scalar model, $C_e/C = V/N$ with $V \sim \chi_N^2$. Put $Z_N = \log(V/N)$. Its moment-generating function near zero is

$$\mathbb{E}e^{uZ_N} = \left(\frac{2}{N}\right)^u \frac{\Gamma(N/2 + u)}{\Gamma(N/2)}. \quad (49)$$

Twice differentiating its logarithm at $u = 0$ yields

$$\text{Var}(Z_N) = \psi_1(N/2) = \frac{2}{N} + O(N^{-2}), \quad (50)$$

where ψ_1 is the second derivative of $\log \Gamma$, called the trigamma function. For example its series $\psi_1(a) = \sum_{j=0}^{\infty} (a+j)^{-2}$ and an integral comparison give $\psi_1(a) = a^{-1} + O(a^{-2})$. This motivates a *standard-deviation* scale $\sigma = \kappa\sqrt{2/N}$, not κ/N . The constant κ includes the time profile of the interpolation variance, $\sigma^2 t(1-t)^2$, and must be chosen numerically. This is a scale comparison, not an optimality theorem for posterior learning.

The same per-coordinate scale arises near an isotropic covariance at fixed d . Set $C = cI_d$ and $E_N = C_e/c - I_d$. Gaussian sampling gives $\sqrt{N} \text{svec}(E_N) \Rightarrow \mathcal{N}(0, 2I_p)$, $p = d(d+1)/2$, using the Frobenius normalization of svec . Since $\log C_e - \log C = E_N + O_P(N^{-1})$, it follows that

$$\sqrt{N} \text{svec}(\log C_e - \log C) \Rightarrow \mathcal{N}(0, 2I_p). \quad (51)$$

This fixed-dimensional local limit does not make matrix-log errors isotropic for arbitrary C , or remain uniform as d/N stays positive. Its leading total squared coordinate scale is proportional to $2p/N$, not $2/N$.

For an RI prior, the observed log eigenvalues do not obey independent scalar chi-square channels. Gaps, repetitions, and d/N affect their noise, so the exact spectral simulator in (16) is the appropriate calibration tool. Re-corruption of draws from the current prior can assess this scale without clean draws from π_* . Smaller σ is not automatically better: the conditional approximation bounds contain $1/\sigma^2$, and both regression and numerical integration change with the scale. The ideal EM theorems are independent of this numerical tuning.

B.7 NUMERICAL SAMPLING

The main implementation integrates the Föllmer SDE (7). More generally, once a compatible fitted drift $\hat{b}^g$ and noise coefficient g have been chosen, Euler–Maruyama uses a grid $0 = t_0 < \dots < t_J = 1$ and independent $z_j \sim \mathcal{N}(0, I_p)$:

$$Y_{j+1} = Y_j + \hat{b}_{t_j}^g(Y_j, Y_e)\Delta t_j + g_{t_j}\sqrt{\Delta t_j} z_j, \quad Y_0 = Y_e. \quad (52)$$

The observation Y_e stays fixed throughout a path; independent paths use independent Gaussian draws. Decode only the endpoint, using Φ^{-1} for full matrices and sorted exponentiation for spectra. The base coefficients are $(\hat{b}, \sigma(1-t))$; the Föllmer coefficients are $((1+t)\hat{b} - y + Y_e, \sigma\sqrt{1-t^2})$. The numerical error term in the supplementary reconstruction bounds concerns the difference between this discrete endpoint law and the continuous-time law. Choosing a step size does not itself certify that error.

C PROOFS OF IDENTIFIABILITY AND NON-IDENTIFIABILITY

C.1 IDENTIFICATION FOR $N \geq d$

For a finite signed measure π on $\mathbb{S}_{++}^d$, define its Laplace transform at $B \succeq 0$ by $\mathcal{L}\pi(B) = \int e^{-\text{tr}(BC)}\pi(dC)$. For a probability prior, the joint characteristic function of the N observations is

$$\varphi_\pi(t_1, \dots, t_N) = \mathbb{E} \exp \left(i \sum_{j=1}^N t_j^T x_j \right) = \mathcal{L}\pi \left(\frac{1}{2} \sum_{j=1}^N t_j t_j^T \right). \quad (53)$$

Every PSD matrix of rank at most N has the displayed representation. If $K_N \pi_1 = K_N \pi_2$, sufficiency (27) implies equality of the raw-sample laws and hence equality of the transforms on this set. For $N \geq d$ the transforms therefore agree for every $B \succ 0$.

For completeness, uniqueness here requires no moment condition on the prior. The transform extends holomorphically to the tube $\{B + iT : B \succ 0, T \in \mathbb{S}^d\}$. Differentiation is justified locally by exponential damping, since $C \succ 0$ and $\|C\|_F \leq \text{tr} C$. Equality on the real open cone implies equality throughout the tube by the identity theorem. Fixing $B_0 \succ 0$, the values at $B_0 + iT$ are the Fourier transforms of the finite tilted measures $e^{-\text{tr}(B_0 C)} \pi_j(dC)$. Fourier uniqueness identifies these measures; the strictly positive tilting factor then identifies π_1 and π_2 . This proves the positive assertion of Theorem 2 for all Borel probability priors.

C.2 SMOOTH RI COUNTEREXAMPLES FOR $N < d$

Let $r = N + 1 \leq d$ and let $e_r(B)$ be the r th elementary symmetric polynomial of the eigenvalues of B . It is a conjugation-invariant homogeneous polynomial, equal to the sum of the principal minors of order r . In particular,

$$e_r(B) = 0 \quad \text{if } \text{rank}(B) \leq N, \quad e_r(tI_d) = \binom{d}{r} t^r > 0 \quad (t > 0). \quad (54)$$

Use the trace inner product on $\mathbb{S}^d$. The matrix gradient has diagonal entries $\partial/\partial C_{ii}$ and off-diagonal entries $\frac{1}{2}\partial/\partial C_{ij}$ in upper-triangular coordinates. This convention ensures $\nabla_C e^{-\text{tr}(BC)} = -B e^{-\text{tr}(BC)}$.

Choose a nonzero, nonnegative, RI function $\vartheta \in C_c^\infty(\mathbb{S}_{++}^d)$. For example, a smooth radial bump around aI_d with radius less than a has these properties. Define

$$f(C) = e_r(\nabla_C) \vartheta(C). \quad (55)$$

This is smooth, compactly supported inside $\mathbb{S}_{++}^d$, and RI because replacing the derivatives in this constant-coefficient operator by a matrix B gives the conjugation-invariant polynomial $e_r(B)$. This polynomial is the operator's symbol. Extending ϑ by zero to $\mathbb{S}^d$ introduces no boundary terms. Repeated integration by parts gives, for $B \succeq 0$,

$$\mathcal{L}f(B) = e_r(B) \mathcal{L}\vartheta(B). \quad (56)$$

Indeed, the sign from integration by parts cancels the sign obtained by differentiating the exponential. Thus $\int f(C) dC = 0$ and $\mathcal{L}f(B) = 0$ whenever $\text{rank}(B) \leq N$. Also $f \not\equiv 0$: at $B = tI_d$, the right-hand side of (56) is strictly positive.

Choose a smooth compactly supported RI probability density p_0 whose support lies inside $\mathbb{S}_{++}^d$ and which is strictly positive on $\text{supp} f$. Such a density is obtained from a slightly larger radial bump. By compactness, $\min_{\text{supp} f} p_0 > 0$ and $\|f\|_\infty < \infty$. Hence for sufficiently small $\varepsilon > 0$,

$$p_+(C) = p_0(C) + \varepsilon f(C), \quad p_-(C) = p_0(C) - \varepsilon f(C) \quad (57)$$

are distinct nonnegative densities of total mass one. Their supports are compact and bounded away from singular matrices. Equations (53) and (56) show that the two raw-sample characteristic functions agree for every $(t_1, \dots, t_N)$. The raw laws, and therefore their empirical-covariance laws, coincide. This proves the negative assertion, including its restriction to smooth RI priors.

The common observation distribution cannot identify the corresponding posterior either. If both posterior kernels coincided almost everywhere under that distribution, integrating them would give equal priors, contradicting $p_+ \neq p_-$. This is an obstruction to full posterior learning, not merely to a chosen representation of the prior.

C.3 A TWO-DIMENSIONAL EXAMPLE

Let $d = 2$, $N = 1$, and $a, b > 0$ with $a \neq b$. Compare the RI priors

$$\begin{aligned} C^{(1)} &= R \operatorname{diag}(a, b) R^T, & R &\sim H, \\ C^{(2)} &= c I_2, & c &= a \cos^2 \Theta + b \sin^2 \Theta, \quad \Theta \sim \operatorname{Unif}[0, 2\pi]. \end{aligned} \quad (58)$$

They are distinct: one has the fixed unequal eigenvalues (a, b) , while the other has two equal random eigenvalues. For any $t \in \mathbb{R}^2$, rotational averaging gives

$$\mathbb{E} \exp\left(-\frac{1}{2} t^T C^{(1)} t\right) = \mathbb{E} \exp\left(-\frac{1}{2} \|t\|_2^2 c\right) = \mathbb{E} \exp\left(-\frac{1}{2} t^T C^{(2)} t\right). \quad (59)$$

Thus one Gaussian observation has the same law under both models, as does $C_e = xx^T$. This example uses singular prior measures for simplicity; the preceding construction establishes the stronger result for smooth full-dimensional densities.

C.4 KNOWN-BASIS DIAGONAL PRIORS

Proposition 7 (Identification of a known-basis diagonal prior). *If $C = \operatorname{diag}(c)$ in a fixed known basis, then $c_e = ((C_e)_{11}, \dots, (C_e)_{dd})$ contains all information about c present in C_e . Its law identifies any joint distribution on $c \in (0, \infty)^d$ for every $N \geq 1$.*

Proof and forward simulator. For $C = \operatorname{diag}(c)$, the part of the raw Gaussian likelihood depending on c is

$$\prod_{j=1}^d c_j^{-N/2} \exp\left[-\frac{N(C_e)_{jj}}{2c_j}\right]. \quad (60)$$

It depends only on $\operatorname{diag} C_e$, which is therefore sufficient for c even when the full empirical covariance is singular. Write $x_{ij} = \sqrt{c_j} z_{ij}$ with all z_{ij} independent standard normals, independent of c . Then $(C_e)_{jj} = c_j V_j / N$, where $V_j = \sum_i z_{ij}^2$ are independent χ_N^2 variables. For $U_j = \log c_j$ and $Z_j = \log(V_j / N)$,

$$\log(C_e)_{jj} = U_j + Z_j, \quad \mathbb{E} e^{i\omega Z_j} = \left(\frac{2}{N}\right)^{i\omega} \frac{\Gamma(N/2 + i\omega)}{\Gamma(N/2)}. \quad (61)$$

The characteristic function of the observed log-diagonal equals the characteristic function of U multiplied by a product of nonzero factors. Division identifies the characteristic function of U , proving Proposition 7. For fixed N , the Gamma function’s large-imaginary-argument asymptotics give

$$\left| \left(\frac{2}{N}\right)^{i\omega} \frac{\Gamma(N/2 + i\omega)}{\Gamma(N/2)} \right| \asymp |\omega|^{N/2-1/2} e^{-\pi|\omega|/2}, \quad |\omega| \rightarrow \infty. \quad (62)$$

Thus high-frequency features of the log-prior distribution have exponentially reduced influence on the observed law. This proof concerns arbitrary joint laws of U , not only independent coordinates.

D POPULATION EM: UPDATE, DESCENT, AND CONVERGENCE

D.1 THE PRIOR-DISTRIBUTION UPDATE

For a current prior π , write $\mu = K_N \pi$ and let $w = d\mu_*/d\mu$. The channel likelihoods in (25) are strictly positive, so all observation mixtures are equivalent to κ_N ; w is therefore defined almost everywhere. Bayes’ formula gives

$$\frac{d\mathcal{T}(\pi)}{d\pi}(C) = r_\pi(C), \quad r_\pi(C) = \int w(C_e) K_N(dC_e | C). \quad (63)$$

The right-hand side integrates to one against π .

To identify this with the unrestricted EM step, let a candidate prior have density p relative to a common dominating measure on the permitted latent space. The part of the complete-data EM surrogate depending on p is

$$\int \int \log p(C) Q_\pi(dC | C_e) \mu_\star(dC_e) = \int \log p(C) \mathcal{T}(\pi)(dC). \quad (64)$$

Its maximizer is the density of $\mathcal{T}(\pi)$ whenever the cross-entropy expression is finite; equivalently, it minimizes $\text{KL}(\mathcal{T}(\pi) \| p)$. The measure-valued posterior average (4) remains meaningful beyond a global density parametrization. This is the latent-distribution EM update of Aubin-Frankowski et al. (2022). Conditional synthetic training implements the next posterior calculation and is not a different maximizer of this surrogate.

D.2 PROOF OF THEOREM 3

Define joint laws

$$\begin{aligned} J_\star(dC, dC_e) &= \pi_\star(dC) K_N(dC_e | C), \\ J_k(dC, dC_e) &= \pi_k(dC) K_N(dC_e | C), \\ H_k(dC, dC_e) &= Q_k(dC | C_e) \mu_\star(dC_e). \end{aligned} \quad (65)$$

Because $J_\star$ and J_k have the same forward conditional, $\text{KL}(J_\star \| J_k) = D_k$. Disintegrating them by C_e instead proves (9). The C -marginals of $J_\star$ and H_k are $\pi_\star$ and π_{k+1} , respectively, and

$$\text{KL}(J_\star \| H_k) = \int \text{KL}(Q_\star(\cdot | C_e) \| Q_k(\cdot | C_e)) \mu_\star(dC_e) = D_k - F_k. \quad (66)$$

Data processing yields $D_{k+1} \leq D_k - F_k$. Starting with finite D_0 ensures that all these quantities are finite at every iteration.

For observation-space monotonicity, put $r_k = d\pi_{k+1}/d\pi_k$. Bayes' formula gives

$$\frac{m_{\pi_{k+1}}(C_e)}{m_{\pi_k}(C_e)} = \int r_k(C) Q_k(dC | C_e). \quad (67)$$

Jensen's inequality then implies

$$\begin{aligned} F_k - F_{k+1} &= \int \log \frac{m_{\pi_{k+1}}(C_e)}{m_{\pi_k}(C_e)} \mu_\star(dC_e) \\ &\geq \int \int \log r_k(C) Q_k(dC | C_e) \mu_\star(dC_e) = \text{KL}(\pi_{k+1} \| \pi_k). \end{aligned} \quad (68)$$

The latter KL is finite because $\text{KL}(H_k \| J_k) = F_k$ and marginalization decreases KL. Standard truncation justifies Jensen if individual log terms are not initially bounded. Summing $D_{k+1} \leq D_k - F_k$ gives $\sum_{j=0}^k F_j \leq D_0$, and monotonicity gives $(k+1)F_k \leq D_0$.

D.3 PROOF OF COROLLARY 4

For compact-support recovery, let $\mathcal{K} \subset \mathbb{S}_{++}^d$ be the common compact set supporting π_0 and $\pi_\star$. Support preservation keeps every π_k in $\mathcal{K}$. Probability measures on $\mathcal{K}$ are weakly sequentially compact. The channel is weakly continuous there: for a bounded continuous function f of the observation,

$$C \mapsto \mathbb{E}_Z f\left(C^{1/2} Z Z^T C^{1/2} / N\right) \quad (69)$$

is continuous by continuity of the positive square root and dominated convergence. Any subsequential weak limit π of π_k therefore satisfies $K_N \pi_k \Rightarrow K_N \pi$ along the subsequence. But $F_k \rightarrow 0$ implies $K_N \pi_k \rightarrow \mu_\star$ in total variation by Pinsker's inequality. Thus $K_N \pi = \mu_\star$. For $N \geq d$, Theorem 2 identifies $\pi = \pi_\star$, so every subsequential limit agrees. The same proof applies to any common compact class on which the channel is injective. This proves $\pi_k \Rightarrow \pi_\star$ without a latent-KL conclusion.

To obtain the conditional statement, fix an admissible observed matrix C_e . The function $\ell_{C_e}(C) = k_N(C_e | C)$ in (25) is continuous, strictly positive, and bounded above and away from zero on $\mathcal{K}$. For every continuous f on $\mathcal{K}$, weak prior convergence therefore gives

$$\int f(C) Q_k(dC | C_e) = \frac{\int f(C) \ell_{C_e}(C) \pi_k(dC)}{\int \ell_{C_e}(C) \pi_k(dC)} \rightarrow \frac{\int f(C) \ell_{C_e}(C) \pi_\star(dC)}{\int \ell_{C_e}(C) \pi_\star(dC)}. \quad (70)$$

The denominator has a strictly positive limit. The likelihood-defined versions of the conditional laws thus converge weakly for every such C_e , and in particular for μ_* -almost every observation. This argument extends the compact-prior convergence proof to the posterior itself; it does not show convergence for all measurable events or in KL. For any fixed bounded continuous test function, dominated convergence also yields convergence of its conditional-expectation discrepancy averaged over μ_* . Since covariance and precision entries are bounded continuous on $\mathcal{K}$, their posterior means converge under these same compactness assumptions.

For the stronger statement, the inverse-stability premise gives $D_{k+1} \leq D_k - F_k \leq (1 - \chi^{-1})D_k$, and induction gives the first part of (12). The posterior identity in Theorem 3 gives the second part because $0 \leq D_k - F_k \leq D_k$. For $\chi = 1$ the first update is already exact; the displayed geometric convention at $k = 0$ is understood as the initial bound D_0 . This inverse-stability condition is not proved for unrestricted Wishart mixtures and is not inferred from injectivity. It is a separate sufficient hypothesis on the iterates and model class.

D.4 SUPPORT RESTRICTION AND NON-OPTIMAL FIXED POINTS

Equation (63) proves $\pi_{k+1} \ll \pi_k$. If $\pi_k = \delta_{C_0}$, then $Q_k(\cdot | C_e) = \delta_{C_0}$ for every admissible observation, so $\mathcal{T}(\pi_k) = \pi_k$ even when it does not explain μ_* . These fixed points do not contradict Theorem 3: finite $\text{KL}(\pi_* || \delta_{C_0})$ would force the truth itself to be that point mass. Conversely, a continuous initial density can have full topological support and still fail to dominate an atomic truth. These distinctions are important when interpreting particle implementations and nominally expressive generative models.

E LOCAL INFORMATION AND TRACTABLE MODES

E.1 LINEARIZATION IN A SPECIFIED NORM

Under the true joint law define

$$A_N h(C_e) = \mathbb{E}[h(C) | C_e], \quad A_N^* g(C) = \mathbb{E}[g(C_e) | C]. \quad (71)$$

For a bounded perturbation $\pi_h = (1+h)\pi_*$ with $\mathbb{E}_{\pi_*} h = 0$ and small $\|h\|_\infty$, the following calculation proves

$$\frac{d\mathcal{T}(\pi_h)}{d\pi_*} - 1 = (\text{Id} - A_N^* A_N)h + O(\|h\|_\infty^2). \quad (72)$$

Write $A = A_N$. For bounded centered h with $\delta = \|h\|_\infty < 1$, the observation law of $\pi_h = (1+h)\pi_*$ has density $1 + Ah$ relative to μ_* . Substituting this into (63) gives

$$\frac{d\mathcal{T}(\pi_h)}{d\pi_*} = (1+h)A_N^* \left[\frac{1}{1 + A_N h} \right]. \quad (73)$$

Using $(1+u)^{-1} = 1 - u + u^2/(1+u)$ gives

$$\frac{d\mathcal{T}(\pi_h)}{d\pi_*} - 1 = (\text{Id} - A^* A)h - hA^* Ah + (1+h)A^* \left[\frac{(Ah)^2}{1 + Ah} \right]. \quad (74)$$

Conditional expectation contracts L^∞ , so the last two terms have norm at most $2\delta^2/(1-\delta)$. This proves the L^∞ expansion in (72). The derivative $\text{Id} - A^* A$ extends continuously to $L_0^2(\pi_*)$, where A^* is the Hilbert adjoint because

$$\mathbb{E}[(Ah)(C_e)g(C_e)] = \mathbb{E}[h(C)g(C_e)] = \mathbb{E}[h(C)(A^*g)(C)]. \quad (75)$$

Conditional Jensen yields $\|Ah\|_2 \leq \|h\|_2$, hence $0 \leq A^* A \leq \text{Id}$. The bounded-perturbation proof is not a claim of Fréchet differentiability in the unrestricted L^2 topology. If bounded nonzero h satisfies $Ah = 0$, the positive priors $(1 \pm \varepsilon h)\pi_*$ have exactly the same observation law for sufficiently small ε .

E.2 MONOTONICITY WITH THE NUMBER OF OBSERVATIONS

Couple all sample sizes through one covariance $C \sim \pi_*$ and an infinite conditionally iid sequence $x_1, x_2, \dots$. For $\mathcal{F}_N = \sigma(x_1, \dots, x_N)$ and centered $h \in L^2(\pi_*)$, sufficiency implies

$$\mathbb{E}[h(C) | \mathcal{F}_N] = \mathbb{E}[h(C) | C_{e,N}] = A_N h(C_{e,N}). \quad (76)$$

Orthogonal projection onto nested sigma-fields yields

$$\|A_N h\|_2^2 \leq \|A_{N+1} h\|_2^2 \leq \|h\|_2^2. \quad (77)$$

Conditional strong laws imply $C_{e,N} \rightarrow C$ almost surely, so $h(C)$ is measurable with respect to $\mathcal{F}_\infty = \sigma(\bigcup_N \mathcal{F}_N)$, up to completion. The L^2 martingale convergence theorem therefore gives $A_N h(C_{e,N}) \rightarrow h(C)$ in L^2 . There is no rate uniform in h in this argument.

E.3 AN EXPLICIT SCALAR MODE

Proposition 8 (Inverse-gamma information mode). *Let $d = 1$ and $c \sim \text{InvGamma}(a, b)$ with density proportional to $c^{-a-1}e^{-b/c}$, $a > 2$, $b > 0$. Put $n = N/2$ and $h(c) = c - b/(a - 1)$. Then*

$$A_N^* A_N h = \frac{n}{a + n - 1} h, \quad (\text{Id} - A_N^* A_N) h = \frac{a - 1}{a + n - 1} h. \quad (78)$$

This is one exact information mode, not a worst-case convergence rate. Since h is unbounded, it is an identity for the square-integrable extension of the linear operator, not an application of the bounded-perturbation remainder to h itself.

Proof. Let $n = N/2$. The scalar channel has likelihood proportional to $c^{-n}e^{-nC_e/c}$ as a function of c . Multiplying by $c^{-a-1}e^{-b/c}$ yields

$$c \mid C_e \sim \text{InvGamma}(a + n, b + nC_e), \quad \mathbb{E}[c \mid C_e] = \frac{b + nC_e}{a + n - 1}. \quad (79)$$

With $\bar{c} = b/(a - 1)$ this gives

$$A_N h(C_e) = \frac{n}{a + n - 1} (C_e - \bar{c}), \quad A_N^* A_N h(c) = \frac{n}{a + n - 1} (\mathbb{E}[C_e \mid c] - \bar{c}) = \frac{n}{a + n - 1} h(c). \quad (80)$$

The condition $a > 2$ guarantees $h \in L^2(\pi_*)$; in fact $\text{Var}(c) = b^2/[(a - 1)^2(a - 2)]$. For a direct numerical test of the derivative, bounded centered truncations of h avoid using the L^∞ remainder outside its domain.

E.4 TENSOR PRODUCTS FOR A FACTORIZED DIAGONAL PRIOR

If the diagonal prior factorizes, both the joint experiment and its posterior factorize over coordinates. On product functions $h(c) = \prod_{j=1}^d h_j(c_j)$, conditional expectation consequently satisfies $Ah = \prod_j A_j h_j$. Thus $A^* A = \bigotimes_j (A_j^* A_j)$ on the Hilbert tensor product. If $A_j^* A_j h_j = \eta_j h_j$, the product is an eigenfunction with eigenvalue $\prod_j \eta_j$. Constant factors have eigenvalue one; modes involving several nonconstant factors may have much smaller information eigenvalues. This tensorization assertion requires a factorized prior and is not assumed in Proposition 7.

F ROTATIONAL INVARIANCE AND FULL POSTERIOR RECONSTRUCTION

An *orthogonal orbit* is the set of matrices $Q\Lambda Q^T$ obtained by rotating a fixed diagonal matrix. Splitting an RI matrix law into its eigenvalue distribution and its conditional distribution on each orbit is called disintegration. The latter conditional distribution is uniform on that orbit. This explains why the spectral reduction differs from an invertible matrix-logarithm transformation.

Theorem 9 (Exact RI reduction). *For an RI prior, $P(\Lambda \in \cdot \mid C_e) = q_\nu(\cdot \mid \Lambda_e)$. If π_1, π_2 are RI with spectral laws ν_1, ν_2 , then*

$$\text{KL}(\pi_1 \parallel \pi_2) = \text{KL}(\nu_1 \parallel \nu_2), \quad \text{KL}(K_N \pi_1 \parallel K_N \pi_2) = \text{KL}(K_N^{\text{spec}} \nu_1 \parallel K_N^{\text{spec}} \nu_2). \quad (81)$$

For RI truth and initialization, exact population EM preserves RI and equals spectral posterior averaging. The prior-identification threshold remains $N \geq d$ within RI priors.

F.1 PROOF OF THEOREM 9

For an ordered positive spectrum Λ , let $\mathcal{H}_\Lambda$ be the pushforward of normalized Haar measure under $U \mapsto U\Lambda U^T$. Every RI prior disintegrates as $\pi = \int \mathcal{H}_\Lambda \nu(d\Lambda)$; this can be seen by averaging any conditional law over the orthogonal group. The kernel $\mathcal{H}_\Lambda$ is common to every RI prior, including at repeated eigenvalues. Because spectra of distinct orbits differ, the KL chain rule gives

$$\text{KL}(\pi_1 \| \pi_2) = \text{KL}(\nu_1 \| \nu_2) + \int \text{KL}(\mathcal{H}_\Lambda \| \mathcal{H}_\Lambda) \nu_1(d\Lambda) = \text{KL}(\nu_1 \| \nu_2). \quad (82)$$

This identity also holds with value $+\infty$ by absolute-continuity considerations. Orthogonal equivariance of the Gaussian channel makes every observation mixture RI. Disintegration on its PSD orbits, including singular ones, gives the second equality in (81).

Conditional on Λ , the spectrum of $C^{1/2} Z Z^T C^{1/2} / N$ is independent of the externally chosen eigenbasis, by orthogonal invariance of Z . This proves (16). To compute the posterior, represent the prior with latent independent $(\Lambda, U) \sim \nu \otimes H$. For a fixed observation choose any orthogonal diagonalization $C_e = U_e \Lambda_e U_e^T$ and set $R = U_e^T U$. Substitution into (25) yields (14), after dropping the factor $\exp(N \text{tr } C_e / 2)$ that is constant in (Λ, R) . The formula is valid for every $N \geq 1$. It depends only on Λ_e , proving spectral sufficiency.

Bayesian equivariance implies $Q_\pi(Q dC Q^T | Q C_e Q^T) = Q_\pi(dC | C_e)$ in pushforward notation. Averaging against the RI law μ_* shows that $\mathcal{T}(\pi)$ is RI. Taking the spectrum of this posterior average gives exactly the spectral EM update in Theorem 9. The reduction concerns the full joint law of the ordered diagonal spectrum, not merely a pooled one-dimensional spectral histogram. The $N \geq d$ identification result follows by lifting spectral laws to RI matrix laws and using Theorem 2; the negative construction in Appendix C is already RI. This proves the threshold assertion in Theorem 9.

F.2 FULL POSTERIOR COMPLETION: PROOF OF THEOREM 5

Fix an observation $C_e = U_e \Lambda_e U_e^T$, and write $\mathcal{O}_{\Lambda, C_e}$ for the conditional law of $C = U_e R \Lambda R^T U_e^T$ when $R \sim H_N(\cdot | \Lambda, \Lambda_e)$. This is a probability kernel supported on matrices with spectrum Λ . It is independent of the spectral prior. Thus

$$Q_*(dC | C_e) = \int \mathcal{O}_{\Lambda, C_e}(dC) q_*(d\Lambda | \Lambda_e), \quad \hat{Q}^H(dC | C_e) = \int \mathcal{O}_{\Lambda, C_e}(dC) \hat{q}(d\Lambda | \Lambda_e). \quad (83)$$

The mapping $C \mapsto (\text{spec } C, C)$ retains all information about C . In these joint coordinates the two laws have different spectral marginals but identical conditional matrix laws. The chain rule for relative entropy therefore proves (21), including infinite values. No unique eigenvector frame is needed. Repeated population eigenvalues merely make the representation by R nonunique; the conditional matrix law remains well defined. A different empirical eigenbasis within a tied eigenspace changes the representation of R but not the exact matrix law.

Exact completion also preserves total variation:

$$\text{TV}(Q_*, \hat{Q}^H) = \text{TV}(q_*, \hat{q}). \quad (84)$$

One inequality follows because applying a common probability kernel cannot increase total variation. The other follows by extracting the ordered spectrum from the matrix. Equivalently, the conditional matrix laws for distinct spectra have disjoint spectral supports.

Proposition 10 (Finite orientation-sampling error). *Fix $C_e = U_e \Lambda_e U_e^T$ and a spectral law $\hat{q}(\cdot | \Lambda_e)$. Let $\hat{Q}^H$ and $\hat{Q}_m$ be its completions using H_N and $\hat{H}_m$, respectively. Write $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$ for total variation, the largest discrepancy in event probabilities, and set $\varepsilon_m(\Lambda, \Lambda_e) = \text{TV}(H_N(\cdot | \Lambda, \Lambda_e), \hat{H}_m(\cdot | \Lambda, \Lambda_e))$. Then*

$$\text{TV}(Q_*, \hat{Q}_m) \leq \text{TV}(q_*, \hat{q}) + \int \varepsilon_m(\Lambda, \Lambda_e) \hat{q}(d\Lambda | \Lambda_e). \quad (85)$$

No uniqueness of an eigenvector frame is required, so this includes repeated eigenvalues.

Proof. Let $\hat{\mathcal{O}}_{\Lambda, C_e}^m$ be the matrix law obtained from $\hat{H}_m$. Deterministic reconstruction gives

$$\text{TV}(\mathcal{O}_{\Lambda, C_e}, \hat{\mathcal{O}}_{\Lambda, C_e}^m) \leq \varepsilon_m(\Lambda, \Lambda_e). \quad (86)$$

Insert the exact completion $\hat{Q}^H$ between Q_* and $\hat{Q}_m$, apply (84), and average the preceding conditional bound under $\hat{q}$. This proves (85). The orientation error is averaged under the generated spectral law, not the unknown true spectral posterior. On matrix space the orientation bound may be conservative because sign flips and rotations within repeated population eigenspaces produce the same matrix.

If instead conditional orientation KL is available, the same chain rule at the augmented (Λ, R) level and deterministic reconstruction give the alternative inequality

$$\begin{aligned} \text{KL}(Q_* \| \hat{Q}_m) &\leq \text{KL}(q_* \| \hat{q}) \\ &+ \int \text{KL}(H_N(\cdot | \Lambda, \Lambda_e) \| \hat{H}_m(\cdot | \Lambda, \Lambda_e)) q_*(d\Lambda | \Lambda_e). \end{aligned} \quad (87)$$

For finite MCMC runs without a full-dimensional transition component, initialization at a point can give a singular output and an infinite conditional KL. An atomic component by itself does not force this KL to be infinite if the output also has a sufficiently positive density. The Haar-mixture kernel below supplies such a density after a step, but the TV formulation also applies to other samplers. The total-variation formulation does not require an everywhere positive approximation density.

F.3 PROOF OF COROLLARY 6

Lift ν_k to the RI prior $\pi_k = \int \mathcal{H}_\Lambda \nu_k(d\Lambda)$. Theorem 9 shows that these are precisely the full-matrix population EM iterates from the RI initialization, and that their D_k, F_k equal D_k^ν, F_k^ν . Exact orientation completion of q_{ν_k} is the posterior of π_k , so $Q_k^H = Q_{\pi_k}$. Theorem 3 proves (22).

If the initial and true spectra belong to a common compact set of positive ordered diagonal matrices, their eigenvalues have common positive lower and finite upper bounds. Its orthogonal rotations form a compact subset of $\mathbb{S}_{++}^d$. For $N \geq d$, Corollary 4 yields weak convergence of the lifted priors and their matrix posteriors. Taking spectra gives $\nu_k \Rightarrow \nu_*$. This proof covers repeated eigenvalues at the distribution level; application through an SI still requires the stated sampler hypotheses. Under $D_k^\nu \leq \chi F_k^\nu$, the stronger averaged-KL rate follows from (12). Exact orientation completion does not add an approximation term. None of these conclusions claims that a finite run of orientation MCMC has already reached its target.

F.4 A CONCRETE ORIENTATION MCMC KERNEL AND ITS GUARANTEE

The following is a specified baseline sampler for Algorithm 2. It is not claimed to be optimal among orthogonal-matrix samplers. The target in (15) belongs to the matrix Bingham family studied by Hoff (2009); the simple mixture below is chosen so its invariance and convergence have a short proof.

Fix $\lambda_i > 0$ and $\lambda_{e,i} \geq 0$, and set $\mathcal{V}(R) = \mathcal{V}_{\Lambda, \Lambda_e}(R)$. Choose $\omega \in (0, 1]$ and a symmetric angle distribution, such as a centered normal with fixed variance. At each step, independently choose one of two proposals. With probability $1 - \omega$, pick $i < j$ uniformly and propose $R' = RG_{ij}(\theta)$, where the (i, j) block of $G_{ij}(\theta)$ is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (88)$$

With probability ω , propose a fresh independent Haar matrix R' . For either proposal, accept with probability (19); otherwise retain R . For $d = 1$ the local proposal can be the identity. A Haar draw on $O(d)$ can be obtained by QR decomposition of an independent Gaussian matrix, with diagonal-sign correction of the triangular factor. This samples both determinant components. Plane rotations alone stay in the initial determinant component, although that sign itself does not affect the reconstructed covariance.

Proposition 11 (Validity and a conservative fixed-spectrum mixing bound). *Let P_{Λ, Λ_e} be this mixture kernel and let H_N denote its target at the fixed pair (Λ, Λ_e) . Define*

$$\Delta_{\Lambda, \Lambda_e} = \max_{R \in O(d)} \mathcal{V}(R) - \min_{R \in O(d)} \mathcal{V}(R) < \infty. \quad (89)$$

Then P_{Λ, Λ_e} is reversible with invariant law H_N , and for any starting distribution η and integer $m \geq 1$,

$$\text{TV}(\eta P_{\Lambda, \Lambda_e}^m, H_N) \leq (1 - \omega e^{-\Delta_{\Lambda, \Lambda_e}})^m. \quad (90)$$

Proof. Haar measure is invariant under right multiplication. Since θ and $-\theta$ have the same law and $G_{ij}(\theta)^{-1} = G_{ij}(-\theta)$, the local proposal is symmetric with respect to Haar measure. The independent-Haar proposal is symmetric as well. The Metropolis acceptance ratio consequently gives detailed balance for each component with density $f(R) = e^{-\mathcal{V}(R)}/Z_{\Lambda, \Lambda_e}$ relative to Haar measure, and hence for their mixture.

The compactness of $O(d)$ and continuity of $\mathcal{V}$ give finite extrema. Put $M_{\Lambda, \Lambda_e} = \sup_R f(R)$. Since $Z_{\Lambda, \Lambda_e} \geq e^{-\max \mathcal{V}}$, we have $M_{\Lambda, \Lambda_e} \leq e^{\Delta_{\Lambda, \Lambda_e}}$. For the independent-Haar Metropolis transition and every state R , its accepted part satisfies

$$\min\{1, f(R')/f(R)\} dH(R') \geq M_{\Lambda, \Lambda_e}^{-1} f(R') dH(R') = M_{\Lambda, \Lambda_e}^{-1} H_N(dR'). \quad (91)$$

Thus the full mixture transition has a common component of weight at least $\epsilon = \omega e^{-\Delta_{\Lambda, \Lambda_e}}$ equal to a fresh draw from H_N . Write it as $P_{\Lambda, \Lambda_e} = \epsilon H_N + (1 - \epsilon) \tilde{P}$, where $\tilde{P}$ is a probability kernel preserving H_N when $\epsilon < 1$. Iteration gives (90); if $\epsilon = 1$ the first step is exact. This also proves uniqueness of the invariant law. $\square$

For example,

$$\Delta_{\Lambda, \Lambda_e} \leq \frac{N}{2} \text{tr}(\Lambda_e)(\lambda_{\min}^{-1} - \lambda_{\max}^{-1}), \quad (92)$$

since $\lambda_{\max}^{-1} I_d \preceq R \Lambda^{-1} R^T \preceq \lambda_{\min}^{-1} I_d$. This bound can be extremely large. Proposition 11 is a fixed-pair convergence guarantee, not a practical dimension-uniform rate or a recommendation to use many global proposals. It implies an upper bound on the ε_m term in Proposition 10; integrating it under the generated spectra bounds $\bar{\varepsilon}_m$. For fixed spectral output law, dominated convergence also gives $\bar{\varepsilon}_m \rightarrow 0$ as $m \rightarrow \infty$, without a uniform useful rate. Other valid Bingham samplers can replace this kernel; their finite-run error must then be analyzed or assessed separately.

Cost of local updates and factored output. Let $A = R^T \Lambda_e R$ and $a_i = \lambda_i^{-1}$. For the rotation convention above,

$$\mathcal{V}(R G_{ij}(\theta)) - \mathcal{V}(R) = \frac{N}{2} (a_i - a_j) [(A_{jj} - A_{ii}) \sin^2 \theta + 2A_{ij} \sin \theta \cos \theta]. \quad (93)$$

The acceptance decision uses this two-by-two block. Upon acceptance, update two columns of R and two rows and columns of A ; this costs $O(d)$ with cached matrices. A sweep through all pairs is still $O(d^3)$, and additional sweeps may be needed. Initializing A or proposing an independent Haar matrix has dense cubic cost. These are operation counts, not mixing guarantees.

Retaining (U_e, R, Λ) is a full sample representation: for any vector v ,

$$Cv = U_e R [\Lambda (R^T (U_e^T v))], \quad C^{-1}v = U_e R [\Lambda^{-1} (R^T (U_e^T v))], \quad \log \det C = \sum_i \log \lambda_i. \quad (94)$$

The two vector operations cost $O(d^2)$ with dense factors; forming the full product is unnecessary. Generic full samples still require quadratic storage. The learned spectral state is smaller, but the output orientation generally has $d(d-1)/2$ continuous degrees of freedom. No linear-cost full-matrix sampling claim follows from RI alone.

F.5 POSTERIOR MEAN AND REPEATED EIGENVALUES

Let $M(C_e) = \mathbb{E}[C \mid C_e]$, which exists almost surely under the moment assumption. For any Q with $QC_e Q^T = C_e$, equivariance gives $QM(C_e)Q^T = M(C_e)$. In an empirical eigenbasis, arbitrary coordinate sign flips force all off-diagonal entries of $U_e^T M(C_e) U_e$ to vanish. If C_e has a repeated eigenvalue, arbitrary orthogonal rotations in its eigenspace force the corresponding diagonal block of the mean to be a scalar multiple of the identity. The diagonal entries depend only on Λ_e by equivariance and are precisely $\mathbb{E}[u_{e,i}^T C u_{e,i} \mid \Lambda_e]$. Using the latent orientation representation gives the expression for ξ_i in (24).

For any coefficient vector z , Frobenius orthogonality gives the finite-sample identity

$$\|C - U_e \text{diag}(z) U_e^T\|_F^2 = \|U_e^T C U_e - \text{diag}(\xi)\|_F^2 + \|\xi - z\|_2^2. \quad (95)$$

Thus $z = \xi$ is the unconstrained diagonal oracle in a fixed empirical eigenbasis. In repeated-eigenvalue blocks, this oracle can depend on the chosen basis and need not itself be an equivariant estimator; the posterior mean remains block-scalar and well defined. For $N \geq d$, the Gaussian sample spectrum is simple almost surely, so this distinction disappears with probability one.

F.6 POSTERIOR LEARNING AND LOSS-SPECIFIC BAYES ACTIONS

The spectral SI at iteration k targets $q_{\nu_k}(d\Lambda \mid \Lambda_e)$. This distribution describes uncertainty in the ordered population eigenvalues under the *current* prior. The superscript (r) in $\Lambda^{(r)}$ indexes posterior draws; a hat in $\hat{\Lambda}_L$ denotes decision coefficients instead. Conditional orientation sampling with (14) converts a spectral draw into a matrix posterior draw. Aligning a sampled Λ with U_e omits this orientation and does not perform that conversion.

The choice between Frobenius and Stein loss is a decision-theoretic operation on this posterior, not a choice of the posterior itself. In particular, for $Q = Q_\pi(\cdot \mid C_e)$ define

$$\mathcal{R}_L(A \mid C_e) = \int L(A, C) Q(dC), \quad \hat{C}_L(C_e) = \arg \min_{A \succ 0} \mathcal{R}_L(A \mid C_e). \quad (96)$$

The underlying prior π may be true or learned. All conditional expectations below are with respect to this same Q .

Squared Frobenius loss. Assume $\mathbb{E}[\|C\|_F^2 \mid C_e] < \infty$. With $M_e = \mathbb{E}[C \mid C_e]$, expansion gives

$$\mathcal{R}_F(A \mid C_e) = \mathbb{E}[\|C - M_e\|_F^2 \mid C_e] + \|A - M_e\|_F^2. \quad (97)$$

Since $M_e \succ 0$, it is the unique minimizer. Under RI, the symmetry argument above gives $M_e = U_e \text{diag}(m(\Lambda_e)) U_e^T$. Thus $\hat{\Lambda}_F = \text{diag}(m(\Lambda_e))$, where $m_i(\Lambda_e) = \mathbb{E}[\sum_j \lambda_j R_{ij}^2 \mid \Lambda_e]$. This averages covariance entries in the empirical frame; it is not generally the average of the ordered population eigenvalues.

Stein loss and its convention. For

$$L_S(A, C) = \text{tr}(C^{-1}A) - \log \det(C^{-1}A) - d, \quad (98)$$

assume $\mathbb{E}[\|C^{-1}\|_F + |\log \det C| \mid C_e] < \infty$ and set $P_e = \mathbb{E}[C^{-1} \mid C_e] \succ 0$. Then

$$\mathcal{R}_S(A \mid C_e) = \text{tr}(P_e A) - \log \det A + \mathbb{E}[\log \det C \mid C_e] - d. \quad (99)$$

Its derivative is $P_e - A^{-1}$ and $-\log \det A$ is strictly convex on the SPD cone, so the unique minimizer is $\hat{C}_S = P_e^{-1}$. Equivalently, the excess risk is

$$\mathcal{R}_S(A \mid C_e) - \mathcal{R}_S(P_e^{-1} \mid C_e) = \text{tr}(P_e A) - \log \det(P_e A) - d \geq 0, \quad (100)$$

with equality only at $A = P_e^{-1}$. The inequality follows by diagonalizing $P_e^{1/2} A P_e^{1/2}$ and using $z - \log z - 1 \geq 0$. Under RI, $P_e = U_e \text{diag}(r(\Lambda_e)) U_e^T$, where

$$\zeta_i = u_{e,i}^T C^{-1} u_{e,i} = \sum_j \lambda_j^{-1} R_{ij}^2, \quad r_i(\Lambda_e) = \mathbb{E}[\zeta_i \mid \Lambda_e], \quad (\hat{\Lambda}_S)_{ii} = r_i(\Lambda_e)^{-1}. \quad (101)$$

Inversion occurs *after* averaging the precision. Neither averaging the reciprocals ζ_i^{-1} nor using $\mathbb{E}[\lambda_i \mid \Lambda_e]$ gives these coefficients in general. Invariance under the stabilizer of C_e forces both P_e and P_e^{-1} to be scalar on repeated empirical eigenspaces, just as for M_e .

The reverse convention $L_S^{\text{rev}}(A, C) = \text{tr}(A^{-1}C) - \log \det(A^{-1}C) - d$ instead has Bayes action $\mathbb{E}[C \mid C_e]$ when the corresponding covariance and log-determinant moments are finite. Indeed, its A -dependent risk is $\text{tr}(A^{-1}M_e) + \log \det A$, uniquely minimized at M_e . Naming the convention is therefore necessary: “Stein-optimal posterior mean” is not correct for (98).

Why square-loss drift regression does not select a decision. The minimizer of (6) is $\mathbb{E}[R_t \mid I_t, C'_e]$, a conditional mean of a *drift target*. Proposition 1 shows that integrating the resulting stochastic dynamics produces the entire conditional endpoint law, not its mean. Positive time weights preserve this exact regression minimizer, although weights and the coordinate map can affect finite-capacity approximation. In particular, training in log-covariance coordinates does not make the terminal output $\exp(\mathbb{E}[\log C \mid C_e])$ or select a log-Euclidean point estimate. The same separation holds for supervised conditional interpolation; self-consistency addresses missing clean training pairs, not a missing decision loss.

F.7 COMPUTING ESTIMATES AFTER LEARNING THE SPECTRAL PRIOR

There are two ways to extract a loss-specific estimate without modifying spectral EM. In either case, the target is the Bayes action for the *learned* RI prior. Matching the true-prior Bayes action additionally requires prior/posterior accuracy and suitable moment control; the posterior KL bounds in this paper alone do not control unbounded covariance or precision moments.

Posterior moment calculation. Draw $(\Lambda^{(r)}, R^{(r)})$ from (14), $r = 1, \dots, J$, and compute

$$\hat{m}_i = \frac{1}{J} \sum_{r=1}^J \sum_j \lambda_j^{(r)} (R_{ij}^{(r)})^2, \quad \hat{r}_i = \frac{1}{J} \sum_{r=1}^J \sum_j (\lambda_j^{(r)})^{-1} (R_{ij}^{(r)})^2. \quad (102)$$

Then $U_e \text{diag}(\hat{m}) U_e^T$ and $U_e \text{diag}((\hat{r}_i^{-1})_i) U_e^T$ are consistent Monte Carlo approximations of the two actions under independent exact draws and the stated moments. The reciprocal estimate is generally biased at finite J . Conditional integration over R can replace orientation sampling whenever tractable; means over a repeated empirical eigenspace can be tied to preserve equivariance at finite J . Spectral draws alone directly estimate spectral functionals such as trace and log-determinant, but do not by themselves yield the empirical-frame coefficients in (24) or (101).

A separate supervised shrinkage predictor. After fitting a spectral prior ν , independently generate

$$\Lambda \sim \nu, \quad U \sim H, \quad C = U \Lambda U^T, \quad C'_e \sim K_N(\cdot \mid C), \quad C'_e = U'_e \Lambda'_e U'^T_e. \quad (103)$$

For a positive vector-valued predictor $f_\theta(\Lambda'_e)$, fit

$$\mathcal{J}_L(\theta) = \mathbb{E}_{\pi_\nu, K_N} [L(U'_e \text{diag}(f_\theta(\Lambda'_e)) U'^T_e, C)]. \quad (104)$$

Here π_ν is the RI matrix prior induced by ν . Assume the relevant population risks are finite and the predictor class can represent the conditional minimizers. The entire empirical spectrum is supplied to f_θ , and equal empirical eigenvalues have tied outputs when rotation equivariance is required. Define the known synthetic targets $\xi'_i = (u'_{e,i})^T C u'_{e,i}$ and $\zeta'_i = (u'_{e,i})^T C^{-1} u'_{e,i}$. By (95), the Frobenius objective differs from $\mathbb{E} \sum_i (f_i(\Lambda'_e) - \xi'_i)^2$ by a term independent of f . The Stein objective differs from

$$\mathbb{E} \sum_i (f_i(\Lambda'_e) \zeta'_i - \log f_i(\Lambda'_e)) \quad (105)$$

by a term independent of f . Conditioning on Λ'_e proves that the unrestricted population minimizers satisfy

$$f_i^F(\Lambda_e) = \mathbb{E}_{\pi_\nu} [\xi_i \mid \Lambda_e], \quad f_i^S(\Lambda_e) = (\mathbb{E}_{\pi_\nu} [\zeta_i \mid \Lambda_e])^{-1}. \quad (106)$$

Thus one learned prior supports both estimators, and the chosen downstream loss changes only the decision predictor. Haar sampling of U in (103) is prior sampling, not sampling from the nonuniform posterior orientation law. This route avoids that posterior group-sampling problem entirely when only a point estimate is needed. For simple empirical spectra, one can further simplify by taking $C = \Lambda$ when computing the invariant synthetic features $(\Lambda'_e, \xi', \zeta')$; a common orthogonal rotation changes none of them. For tied eigenspaces, use block-averaged targets with tied predictor outputs.

Neither predictor is fed back as the reconstructed target. Spectral EM re-corrupts draws from $q_{\nu_k}(d\Lambda \mid \Lambda_e)$; replacing them by $f^F(\Lambda_e)$, $f^S(\Lambda_e)$, or ξ defines a different latent distribution and loses the EM interpretation. These coefficients optimize decisions in the observed eigenbasis; they are not population-spectrum samples. This distinction is independent of whether the posterior was learned from supervised pairs or through self-consistency.

F.8 NONLINEAR SHRINKAGE AND AN AVERAGED BAYES COMPARISON

For $d/N \rightarrow q \in (0, 1)$, the Ledoit–Péché expression for asymptotic directional-variance coefficients has the form

$$\begin{aligned} \delta_i^{\text{LP}}(\Lambda_e) &= \frac{\lambda_{e,i}}{|1 - q - q\lambda_{e,i}\check{m}_F(\lambda_{e,i})|^2}, \\ m_F(z) &= \int_{[0,\infty)} \frac{F(du)}{u - z}, \quad \check{m}_F(x) = \lim_{\epsilon \downarrow 0} m_F(x + i\epsilon), \quad x > 0. \end{aligned} \quad (107)$$

Here F is the limiting sample spectral distribution under the assumptions of the random-matrix results, and the displayed integral defines its Stieltjes transform. It is not the raw finite-sample eigenvalue measure. Ledoit–Wolf nonlinear shrinkage estimates the required spectral quantities; see Ledoit & Péché (2011); Ledoit & Wolf (2012). Neither method is defined as the posterior mean of the ordered population eigenvalues under our unknown ensemble prior.

Proposition 12 (An averaged connection to the RI Bayes estimate). *For a square-integrable vector $\delta(\Lambda_e)$ measurable in the sample spectrum, and $m(\Lambda_e) = \mathbb{E}[\xi \mid \Lambda_e]$,*

$$\frac{1}{d} \mathbb{E} \|m(\Lambda_e) - \delta(\Lambda_e)\|_2^2 \leq \frac{1}{d} \mathbb{E} \|\xi - \delta(\Lambda_e)\|_2^2. \quad (108)$$

Thus any averaged squared-error approximation of the directional oracle ξ by LP-type coefficients gives the corresponding approximation of the RI Frobenius Bayes coefficients.

Proof and scope. Conditional expectation is an orthogonal projection in L^2 . Since $m(\Lambda_e) = \mathbb{E}[\xi \mid \Lambda_e]$ and $\delta(\Lambda_e)$ is measurable in Λ_e ,

$$\mathbb{E} \|\xi - \delta(\Lambda_e)\|_2^2 = \mathbb{E} \|\xi - m(\Lambda_e)\|_2^2 + \mathbb{E} \|m(\Lambda_e) - \delta(\Lambda_e)\|_2^2. \quad (109)$$

This proves (108). For a sequence of RI models indexed by d, N , vanishing of the first displayed right-hand risk $d^{-1} \mathbb{E} \|\xi - \delta(\Lambda_e)\|_2^2$ implies vanishing of $d^{-1} \mathbb{E} \|m(\Lambda_e) - \delta(\Lambda_e)\|_2^2$ without further argument. However, that premise must itself be established in the considered random-matrix regime. Weak convergence of a weighted spectral measure alone does not imply it. One must also distinguish a deterministic limiting F , for which (107) defines a spectrum-measurable oracle, from an unobserved task-dependent random limiting law. In the latter situation a coefficient using that hidden law is not necessarily measurable in the observed spectrum, and the proposition cannot be applied without additional conditioning or estimation.

Under RI, the excess Bayes Frobenius risk of the estimator $U_e \text{diag}(\delta(\Lambda_e)) U_e^T$ is exactly $\mathbb{E} \|m(\Lambda_e) - \delta(\Lambda_e)\|_2^2$. This connects coefficient approximation to point-estimation risk, but says nothing about the quality of posterior draws. Sampling $\text{diag}(\xi)$ in the empirical frame discards off-diagonal posterior fluctuations; sampling Λ in that frame discards the relative orientation. Neither generally reproduces the full matrix posterior.

G DRIFT APPROXIMATION, OUTER-UPDATE ERROR, AND FINITE ENSEMBLES

A learned conditional kernel need not equal Q_ρ for any prior ρ under the fixed channel. Nevertheless, its reconstructions define an actual output law. The purpose of this section is to compare that output with the exact EM update of the *prior supplying its synthetic training pairs*, using the regression error of the conditional interpolant. All population-risk quantities below are evaluated at the fitted network; no assumption of exact finite-data optimization is made. The ideal identifiability and EM recovery results do not rely on the bounds in this appendix.

G.1 SUPPLEMENTARY ONE-UPDATE BOUND FROM DRIFT REGRESSION

For a current prior π , define the excess drift error and its Föllmer-weighted integral by

$$e_t = \mathbb{E}_{\pi_{K_N, W}} \|\hat{b}_t(I_t, Y'_e) - b_t^\pi(I_t, Y'_e)\|_2^2, \quad (110)$$

$$\mathcal{E}_\pi^F = \frac{1}{2\sigma^2} \int_0^1 \frac{1+t}{1-t} e_t dt. \quad (111)$$

Conditional-expectation orthogonality gives $\mathcal{L}_\pi(\widehat{b}) - \mathcal{L}_\pi(b^\pi) = \int w(t)e_t dt$. Thus e_t excludes the unavoidable random variation of the regression target. The singular endpoint weights are not automatically controlled by a small unweighted training loss.

Theorem 13 (From drift regression to a reconstruction update). *Assume the endpoint and change-of-drift hypotheses in Appendix B, finite $\mathcal{E}_\pi^F$, and $d\mu_\star/d\mu_\pi \leq B_\pi < \infty$. Let $\widehat{Q}^F$ be the learned continuous-time covariance kernel and $\widehat{\pi}^+ = \int \widehat{Q}^F(\cdot | C_e)\mu_\star(dC_e)$. Then*

$$\int \text{KL}(Q_\pi(\cdot | C_e) \| \widehat{Q}^F(\cdot | C_e))\mu_\pi(dC_e) \leq \mathcal{E}_\pi^F, \quad (112)$$

$$\text{KL}(\mathcal{T}(\pi) \| \widehat{\pi}^+) \leq B_\pi \mathcal{E}_\pi^F. \quad (113)$$

The corresponding result for other admissible noise coefficients is given in Appendix B.

The overlap bound B_π transfers synthetic-training error to real observations; a positive likelihood does not imply such a bound. The proof combines the trajectory calculation of Chen et al. (2024) with posterior averaging (Appendix G).

To include error in the current prior and numerical integration, write $D(\pi) = \text{KL}(\pi_\star \| \pi)$, $F(\pi) = \text{KL}(\mu_\star \| \mu_\pi)$, and let $\widehat{Q}_{\text{num}}^F$ be the numerical endpoint law. Total variation, $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$, measures the largest discrepancy in event probabilities. If $D(\pi) < \infty$ and $\mathbb{E}_{\mu_\star} \text{TV}(\widehat{Q}^F, \widehat{Q}_{\text{num}}^F) \leq \delta_{\text{num}}$, the KL identity and Pinsker’s inequality give

$$\mathbb{E}_{\mu_\star} \text{TV}(Q_\star, \widehat{Q}_{\text{num}}^F) \leq \sqrt{\frac{D(\pi) - F(\pi)}{2}} + \sqrt{\frac{B_\pi \mathcal{E}_\pi^F}{2}} + \delta_{\text{num}}. \quad (114)$$

Kernels in this display are evaluated at the same C_e . The terms separate prior learning, conditional regression, and integration error. They are theoretical quantities, not certificates computable from raw training loss. The bound concerns a current prior; it is not unrestricted finite-data neural convergence. The remaining subsections give finite-iteration tracking and discuss empirical sampling and endpoint control. These results are not needed for the ideal population theorems.

G.2 CHANGING THE OBSERVATION DISTRIBUTION

Proposition 14 (A conditional one-step approximation bound). *Fix a prior π with observation law $\mu = K_N \pi$, exact posterior Q_π , and approximate kernel $\widehat{Q}$. Suppose $d\mu_\star/d\mu \leq R$ almost everywhere and*

$$\varepsilon = \int \text{KL}(Q_\pi(\cdot | C_e) \| \widehat{Q}(\cdot | C_e))\mu(dC_e) < \infty. \quad (115)$$

For $\widehat{\pi}^+ = \int \widehat{Q}(\cdot | C_e)\mu_\star(dC_e)$,

$$\text{KL}(\mathcal{T}(\pi) \| \widehat{\pi}^+) \leq R\varepsilon, \quad \text{TV}(\mathcal{T}(\pi), \widehat{\pi}^+) \leq \sqrt{R\varepsilon/2}, \quad (116)$$

where total variation is $\sup_B |\rho(B) - \rho'(B)|$.

Proof. The two joint laws with common observation marginal $\mu_\star$ and conditional kernels Q_π and $\widehat{Q}$ have KL divergence equal to the $\mu_\star$ -average of the conditional KL. The density-ratio assumption bounds it by $R\varepsilon$. Marginalization and Pinsker’s inequality give the two assertions. $\square$

The density-ratio condition transfers an average conditional error under the synthetic observation law to the real observation law. Common support alone is insufficient. In particular, this lemma does not use a triangle inequality for KL, and cannot be applied by adding its right-hand side directly to the latent KL recursion of Theorem 3.

G.3 PROOF OF THEOREM 13

For the base and Föllmer coefficients with scale σ , define $\mathcal{E}_\pi^{\text{base}} = (2\sigma^2)^{-1} \int_0^1 e_t/(1-t)^2 dt$ and $\mathcal{E}_\pi^F = (2\sigma^2)^{-1} \int_0^1 (1+t)e_t/(1-t) dt$. Write $\mathcal{E}_\pi^g$ for either value when its corresponding noise

coefficient is chosen. The Föllmer value is exactly (111) for the scale $\sigma > 0$ retained in the main implementation. Fix π , an observation C_e , and its encoding Y_e . We prove the statement for general linear-schedule noise scale $\sigma > 0$ and both the base and Föllmer samplers; Theorem 13 is the Föllmer case with this same σ . The coordinate transform Φ is invertible, so endpoint KL in covariance coordinates equals endpoint KL in Y coordinates. The exact and learned base SDEs have common initial state and volatility $\sigma(1-t)$. Under the stated Girsanov conditions, their conditional path KL is

$$\frac{1}{2\sigma^2} \int_0^1 \frac{\mathbb{E}[\|\hat{b}_t(I_t, Y_e) - b_t^\pi(I_t, Y_e)\|_2^2 | C_e]}{(1-t)^2} dt. \quad (117)$$

For the Föllmer variant, (7) gives $\hat{b}_t^F - b_t^{\pi, F} = (1+t)(\hat{b}_t - b_t^\pi)$ and $g_t^2 = \sigma^2(1-t^2)$. The conditional path KL therefore has weight

$$\frac{(1+t)^2}{2\sigma^2(1-t^2)} = \frac{1+t}{2\sigma^2(1-t)}. \quad (118)$$

The exact base and Föllmer SDEs share the conditional interpolant marginals, so both expectations are evaluated using the same synthetic (I_t, Y_e') . Marginalizing to time one and averaging against μ_π gives (112). Proposition 14, with $R = B_\pi$, then yields (113) and the bound $\text{TV}(\mathcal{T}(\pi), \hat{\pi}^+) \leq \sqrt{B_\pi \mathcal{E}_\pi^g/2}$. For spectral sampling, replace matrix variables by their spectral counterparts and the forward channel by K_N^{spec} ; the same argument applies, using data processing at the final sort/decoding step. The known-basis diagonal version is identical with the simulator stated in Appendix C.4. All assertions require finite energy on the full interval and the corresponding endpoint absolute continuity; a calculation only on $[0, 1-\tau]$ controls that intermediate marginal, not automatically the intended covariance posterior.

G.4 PROOF OF THE FULL POSTERIOR-RECONSTRUCTION BOUND

The following argument applies to either noise choice covered by the preceding proof; take $g = g^F$ for the bound stated in this appendix. For a current prior π with $D(\pi) < \infty$, the joint-law calculation from Theorem 3 does not require π to belong to an exact EM trajectory. It gives

$$\int \text{KL}(Q_\star(\cdot | C_e) \| Q_\pi(\cdot | C_e)) \mu_\star(dC_e) = D(\pi) - F(\pi). \quad (119)$$

Conditional Pinsker's inequality and Jensen's inequality for the square root imply

$$\int \text{TV}(Q_\star(\cdot | C_e), Q_\pi(\cdot | C_e)) \mu_\star(dC_e) \leq \sqrt{\frac{D(\pi) - F(\pi)}{2}}. \quad (120)$$

Theorem 13 and the observation-overlap bound give, before marginalizing the latent variable,

$$\int \text{KL}(Q_\pi(\cdot | C_e) \| \hat{Q}^g(\cdot | C_e)) \mu_\star(dC_e) \leq B_\pi \mathcal{E}_\pi^g. \quad (121)$$

Apply Pinsker and Jensen again, and insert Q_π and $\hat{Q}^g$ between $Q_\star$ and $\hat{Q}_{\text{num}}^g$ in the total-variation triangle inequality. The assumed numerical error then proves (114). This is an average over real observations, not a bound for every individual observation. It controls probabilities of posterior events and bounded test functions, not unbounded moments such as arbitrary covariance or precision entries without additional assumptions.

The ratio bound and weighted regression error are assumptions about population laws. They are not estimated by the finite training objective alone. In particular, neither a low empirical regression loss nor a plausible output covariance supplies the first term, which concerns the unknown true prior. The bound describes which errors must be controlled; it is not a directly computable stopping certificate.

G.5 FINITE-ITERATION TRACKING OF THE EXACT MAP

The one-step error can be propagated in a genuine metric. The following elementary stability lemma makes explicit one sufficient route, without assuming that the EM map is contractive.

Lemma 15 (Overlap-dependent stability of posterior averaging). *For priors ρ, π , if $d\mu_\star/d(K_N\rho) \leq R$, then*

$$\text{TV}(\mathcal{T}(\rho), \mathcal{T}(\pi)) \leq 2R \text{TV}(\rho, \pi). \quad (122)$$

Proof. Set $\mu_\rho = K_N \rho$ and $\mu_\pi = K_N \pi$. Inserting the joint law $\mu_\pi(dC_e)Q_\pi(dC | C_e)$ and using the TV triangle inequality gives

$$\begin{aligned} \int \text{TV}(Q_\rho(\cdot | C_e), Q_\pi(\cdot | C_e)) \mu_\rho(dC_e) &\leq \text{TV}(\rho K_N, \pi K_N) + \text{TV}(\mu_\rho, \mu_\pi) \\ &\leq 2 \text{TV}(\rho, \pi). \end{aligned} \quad (123)$$

The first joint TV equals $\text{TV}(\rho, \pi)$ because the latent coordinate is retained and the forward kernel is common; the second decreases under the forward channel. Change measure from μ_ρ to $\mu_\star$ and marginalize the latent output to obtain (122). $\square$

Corollary 16 (Tracking for finitely many learned updates). *Let ρ_k be the priors supplying the synthetic training pairs of successive learned drifts. Suppose Theorem 13 applies at each ρ_k with overlap bound R_k and energy $\mathcal{E}_k$. Let $H_k = \int \hat{Q}_k(\cdot | C_e) \mu_\star(dC_e)$ be the learned continuous-time population output, and suppose $\text{TV}(\rho_{k+1}, H_k) \leq \delta_k$. For the exact reference trajectory $\pi_{k+1} = \mathcal{T}(\pi_k)$ with $\pi_0 = \rho_0$, write $d_k = \text{TV}(\rho_k, \pi_k)$. Then*

$$d_{k+1} \leq 2R_k d_k + \sqrt{R_k \mathcal{E}_k / 2} + \delta_k, \quad (124)$$

and, for each finite K ,

$$d_K \leq \sum_{j=0}^{K-1} \left(\sqrt{R_j \mathcal{E}_j / 2} + \delta_j \right) \prod_{\ell=j+1}^{K-1} 2R_\ell. \quad (125)$$

Proof. Insert H_k , $\mathcal{T}(\rho_k)$, and $\mathcal{T}(\pi_k)$ between ρ_{k+1} and π_{k+1} in the TV triangle inequality. Use the numerical/ensemble premise, Theorem 13, and Lemma 15, respectively. Iterate the resulting recurrence from $d_0 = 0$. $\square$

The overlap bounds can make this estimate conservative: $R_k \geq 1$, so (125) is not a uniform-iteration convergence bound. It does establish approximation of any fixed finite exact trajectory as the stated errors vanish with controlled overlap. If an additional stability estimate gives a factor $r < 1$ in the *same metric* on an invariant model class, the factor $2R_k$ can be replaced by r , yielding a geometric accumulated-error bound. The L^2 linearization in Section 3.4 does not by itself establish such TV contraction.

G.6 ENDPOINT-COMPATIBLE PARAMETRIZATION AND EXCESS RISK

For the linear schedules, the exact endpoint regression is $b_1^\pi(y, Y_e) = y - Y_e$ because Y and W_1 are conditionally independent at $t = 1$. One possible architecture therefore writes

$$\hat{b}_\theta(t, y, Y_e) = y - Y_e + (1 - t)u_\theta(t, y, Y_e). \quad (126)$$

For $t < 1$ let $u_t^\pi = (b_t^\pi - y + Y_e)/(1 - t)$. Whenever this residual is well defined and the relevant expectations are finite, put $e_t^u = \mathbb{E}_{\pi K_N, W} \|u_\theta(t, I_t, Y_e') - u_t^\pi(I_t, Y_e')\|_2^2$. Then

$$e_t = (1 - t)^2 e_t^u, \quad \mathcal{E}_\pi^{\text{base}} = \frac{1}{2\sigma^2} \int_0^1 e_t^u dt, \quad \mathcal{E}_\pi^{\text{F}} = \frac{1}{2\sigma^2} \int_0^1 (1 - t^2) e_t^u dt. \quad (127)$$

Thus a square-integrable residual error gives finite endpoint energies for both samplers. This is a conditional design implication, not a regularity guarantee for every prior. The raw regression loss should be evaluated as in (6), without dividing the random target by $1 - t$. Multiplying the *raw* square loss by the singular weights in (111) can give an infinite objective because its irreducible conditional variance need not vanish. The theory controls weighted excess risk, which must not be confused with that raw objective. Endpoint cutoff, finite-variance time weighting, and the residual parametrization are distinct implementation choices to evaluate.

G.7 FINITE OBSERVATION AND SYNTHETIC ENSEMBLES

For a finite observed ensemble, let $\hat{\mu}_M = M^{-1} \sum_{a=1}^M \delta_{C_{e,a}}$. The empirical objective is

$$\hat{\ell}_M(\pi) = \frac{1}{M} \sum_{a=1}^M \log m_\pi(C_{e,a}), \quad (128)$$

where the reference κ_N is fixed. In this Gaussian experiment the observation law is nonatomic, so $\text{KL}(\hat{\mu}_M \| K_N \pi) = +\infty$. It is therefore incorrect to replace μ_* by $\hat{\mu}_M$ directly inside Theorem 3. Exact finite-data EM instead increases (128), with the same Jensen argument as for population EM after replacing the observation integral by a finite average (when the logarithms are finite). This remains an empirical likelihood statement even if drift regression and conditional sampling are exact. Recovery as M grows, or recovery for a fixed M , requires separate statistical assumptions; neither follows from the population theorem alone. Likewise, the density-ratio premise of Proposition 14 does not hold with an atomic empirical marginal against a continuous synthetic one. Regularized model classes, held-out validation, and early stopping are plausible design choices, not established statistical guarantees in this draft.

In Corollary 16, δ_k can include the discrepancy caused by replacing the population observation average by the finite ensemble and by numerical integration, but no distribution-free rate for this quantity is asserted. Training on finitely many synthetic pairs contributes to the population regression error $\mathcal{E}_k$. These errors may depend on all previous iterates and on the same reused dataset. A finite cached particle population may also create atomic target laws outside the endpoint-KL assumptions. Fresh draws from a continuous implicit prior and a generalization analysis are different from an exact finite-particle EM update; they should not be conflated.

G.8 SUPPLEMENTARY RI RECONSTRUCTION BOUND WITH INCOMPLETE LEARNING

For a current spectral prior ν , put $D_\nu = \text{KL}(\nu_* \| \nu)$ and $F_\nu = \text{KL}(\mu_*^{\text{spec}} \| K_N^{\text{spec}} \nu)$. Assume the spectral version of Theorem 13, with overlap bound B_ν , weighted error $\mathcal{E}_\nu^F$, and average numerical spectral error δ_{spec} . With $\hat{q}_{\text{num}}$ in (20), define

$$\bar{\varepsilon}_m = \int \int \varepsilon_m(\Lambda, \Lambda_e) \hat{q}_{\text{num}}(d\Lambda \mid \Lambda_e) \mu_*^{\text{spec}}(d\Lambda_e). \quad (129)$$

For $D_\nu < \infty$, the resulting full reconstruction satisfies

$$\mathbb{E}_{\mu_*} \text{TV}(Q_*, \hat{Q}_m) \leq \sqrt{\frac{D_\nu - F_\nu}{2}} + \sqrt{\frac{B_\nu \mathcal{E}_\nu^F}{2}} + \delta_{\text{spec}} + \bar{\varepsilon}_m. \quad (130)$$

The first three terms are the supplementary general-case bound applied to spectra. The fourth is solely the cost of finite orientation sampling. The fixed-spectrum MCMC guarantee in Appendix F.4 need not be uniform over dimensions or spectral draws; efficient mixing must be measured rather than inferred from the known density.

Proof. Apply the proof of (114) to the spectral model (ν, K_N^{spec}) . The required exact conditional-sampling statement follows from the same Brownian-interpolant argument in dimension d . Decoding by exponentiation and sorting cannot increase the discrepancy of numerical states; at exact endpoints sorting changes nothing. Under the preceding spectral assumptions,

$$\int \text{TV}(q_*(\cdot \mid \Lambda_e), \hat{q}_{\text{num}}(\cdot \mid \Lambda_e)) \mu_*^{\text{spec}}(d\Lambda_e) \leq \sqrt{\frac{D_\nu - F_\nu}{2}} + \sqrt{\frac{B_\nu \mathcal{E}_\nu^F}{2}} + \delta_{\text{spec}}. \quad (131)$$

Now apply (85) with $\hat{q} = \hat{q}_{\text{num}}$ at each observation. The bound depends on C_e only through Λ_e , and its average is therefore over μ_*^{spec} . Adding the generated-spectrum average of ε_m proves (130). In particular, exact orientation sampling contributes no additional error to the spectral learning theorem. Spectral simulation costs and finite-time MCMC errors do not disappear from the implementation merely because the statistical reduction is exact.

H EXACT-POSTERIOR BENCHMARKS AND PROPOSED NUMERICAL TESTS

This appendix specifies controlled tests for the general and RI reconstruction methods. The benchmark generator knows the covariance prior and can compute its posterior; the self-consistent learner receives only the empirical-covariance ensemble, N , and the forward simulator. Clean matrices and prior parameters are reserved for evaluation, except in a separately identified supervised control. These are proposed experiments, not numerical results. Their purpose is to distinguish recovery of a posterior distribution from recovery of one covariance estimate.

H.1 INVERSE-WISHART EXAMPLES: ONE FAMILY FOR BOTH IMPLEMENTATIONS

We use the following explicit convention for the inverse-Wishart distribution:

$$C \sim \text{IW}_d(\nu_0, \Psi_0), \quad p(C) = \frac{|\Psi_0|^{\nu_0/2}}{2^{\nu_0 d/2} \Gamma_d(\nu_0/2)} |C|^{-(\nu_0+d+1)/2} \exp \left[-\frac{1}{2} \text{tr}(\Psi_0 C^{-1}) \right],$$

$$\Gamma_d(a) = \pi^{d(d-1)/4} \prod_{i=1}^d \Gamma \left(a + \frac{1-i}{2} \right). \quad (132)$$

The density is relative to the independent entries of a symmetric matrix. Here ν_0 is a scalar number of degrees of freedom and $\Psi_0 \succ 0$ is a scale matrix. The law is proper for $\nu_0 > d - 1$. Its usefulness here is conjugacy: multiplying its density by the Gaussian covariance likelihood gives another inverse-Wishart density (Alvarez et al., 2014). In the known-zero-mean model of (1), collecting the powers of $|C|$ and the terms in C^{-1} gives

$$C \mid C_e \sim \text{IW}_d(n, B), \quad n = \nu_0 + N, \quad B = \Psi_0 + N C_e. \quad (133)$$

This formula uses $N C_e = \sum_i x_i x_i^T$, without centering by an estimated mean. It also holds for singular C_e because it follows from the raw Gaussian likelihood and $B \succ 0$. For reconstruction experiments without a restrictive prior assumption, the principal tests nevertheless use $N \geq d$, in accordance with Theorem 2. A known-prior reference posterior below this threshold does not make the unknown-prior problem identifiable.

RI example. Set $\nu_0 = d + 1 + \kappa_0$ and $\Psi_0 = \kappa_0 c_0 I_d$, where $c_0 > 0$ and $\kappa_0 > 2$. The prior is RI and $\mathbb{E}C = c_0 I_d$. Both Algorithms 1 and 2 can be tested against exactly the same full posterior. In particular,

$$\mathbb{E}[C \mid C_e] = \frac{N}{N + \kappa_0} C_e + \frac{\kappa_0}{N + \kappa_0} c_0 I_d. \quad (134)$$

Thus the exact posterior mean is a linear shrinkage rule in this benchmark. Recovering it alone is not evidence that a full posterior has been learned, nor does this example establish a benefit over nonlinear shrinkage for arbitrary population spectra. The posterior still has uncertainty in both eigenvalues and orientations.

Non-RI example. Replace $c_0 I_d$ by a fixed nonspherical $C_0 \succ 0$ and take $\Psi_0 = \kappa_0 C_0$. Keep the directions of C_0 fixed across tasks rather than rotating it independently for every task. This introduces shared directional information that the general method can learn. Formula (134) then holds with C_0 in place of $c_0 I_d$. A controlled choice is $C_0 = Q_0 \text{diag}(c_1, \dots, c_d) Q_0^T$, with Q_0 drawn once and fixed, and c_i geometrically spaced before normalizing their mean to one. Normalize the chosen scale matrix, not the sampled covariances: fixing the trace of each draw would destroy (133).

Moment and convergence assumptions. The second-moment formulas below require $n > d + 3$. Choosing $\kappa_0 > 6$ also supplies finite fourth moments before conditioning, which is useful when estimating the sampling error of second-moment estimates (Matsumoto, 2012). Higher moments must be checked individually. These priors have unbounded support, so the compact-support recovery corollary in the main text does not apply automatically. Neither an explicit reference posterior nor good finite-data performance proves the unrestricted population convergence claim without its additional assumptions.

H.2 INDEPENDENT REFERENCE SAMPLES AND A FULL-MATRIX CHECK

At a fixed C_e , compute a lower-triangular Cholesky factor L_B with $B = L_B L_B^T$. Independently generate a lower-triangular matrix T whose entries satisfy

$$T_{ii} = \sqrt{\chi_{n-i+1}^2}, \quad T_{ij} \sim \mathcal{N}(0, 1) \ (i > j), \quad T_{ij} = 0 \ (i < j), \quad (135)$$

with all displayed random variables mutually independent. The Bartlett construction gives $TT^T \sim W_d(n, I_d)$. Consequently,

$$C^{\text{ref}} = L_B (TT^T)^{-1} L_B^T \sim \text{IW}_d(n, B). \quad (136)$$

In code, solve $TD = L_B^T$ and return $D^T D$ instead of explicitly forming the inverse. This is a reference construction, not a claim of an optimal implementation; direct inverse-Wishart factor samplers are also available (Axen, 2023). Independent reference draws require no MCMC or equilibration period. Diagonalizing independent C^{ref} gives independent reference spectra; under RI these have exactly the desired law $P_\star(d\Lambda \mid \Lambda_e)$.

The same construction supplies a strong diagnostic for *full* posterior samples. Given a generated C , form

$$W = L_B^T C^{-1} L_B, \quad T = \text{chol}(W). \quad (137)$$

For a correct posterior draw, $W \sim W_d(n, I_d)$ and the Cholesky entries have the independent distributions in (135). Test the diagonal squares against their chi-squared distributions and the lower off-diagonal entries against standard normals. Check dependence between entries as well: correct one-dimensional histograms alone do not establish the joint distribution. This transformation is applied separately at each observation, with its own B ; it can detect orientation errors even when the spectrum of C has been reproduced.

H.3 ANALYTIC POSTERIOR QUANTITIES FOR TESTING ACCURACY

Fix C_e and put $r = n - d - 1 > 2$. The following formulas provide checks without a Monte Carlo approximation to the reference moments. First and second inverse-Wishart moments can be obtained from the inverse-Wishart moment identities; see, for example, Matsumoto (2012). With the convention (132), they give

$$M_\star = \mathbb{E}[C \mid C_e] = \frac{B}{r}, \quad \Omega_\star = \mathbb{E}[C^{-1} \mid C_e] = nB^{-1}, \quad (138)$$

$$\text{Cov}(C_{ij}, C_{k\ell} \mid C_e) = \frac{2B_{ij}B_{k\ell} + r(B_{ik}B_{j\ell} + B_{i\ell}B_{jk})}{(r+1)r^2(r-2)}. \quad (139)$$

The precision mean in (138) also follows directly from $C^{-1} \mid C_e \sim W_d(n, B^{-1})$. The corresponding Stein-optimal covariance estimate, for the loss convention in Appendix F.6, is $\Omega_\star^{-1} = B/n$, not $M_\star$.

Covariance and precision means. For J generated draws, estimate these quantities by $\widehat{M} = J^{-1} \sum_j C^{(j)}$ and $\widehat{\Omega} = J^{-1} \sum_j (C^{(j)})^{-1}$. Report

$$e_M = \frac{\|\widehat{M} - M_\star\|_F}{\|M_\star\|_F}, \quad e_\Omega = \frac{\|\widehat{\Omega} - \Omega_\star\|_F}{\|\Omega_\star\|_F}. \quad (140)$$

Also compare entrywise variances and a specified subset of covariances using (139). The full array has $O(d^4)$ entries, so large-dimensional tests should fix a reproducible subset rather than promise to estimate every entry. Mean errors and variance errors answer different questions: a sampler returning only $M_\star$ would pass a first-moment check and fail uncertainty checks.

Directional-variance distributions. For a nonzero direction v fixed conditional on C_e , the scalar $V_v = v^T C v$ has distribution

$$V_v \mid C_e \sim \text{InvGamma}\left(\frac{r+2}{2}, \frac{v^T B v}{2}\right), \quad (141)$$

$$\mathbb{E}[V_v \mid C_e] = \frac{v^T B v}{r}, \quad \text{Var}(V_v \mid C_e) = \frac{2(v^T B v)^2}{r^2(r-2)}.$$

Here the inverse-gamma density is proportional to $x^{-a-1}e^{-b/x}$ for shape a and scale b . To see the distributional statement, rotate and rescale v to a coordinate direction and use the one-dimensional inverse-Wishart marginal (Alvarez et al., 2014). Choose coordinate vectors, empirical eigenvectors, and selected sums $(u_{e,i} + u_{e,j})/\sqrt{2}$. The sums involve off-diagonal covariance entries and therefore test more than variances along empirical axes. Directions may depend on the observation but not on the generated posterior sample being tested.

Let F_{v,C_e} be the exact cumulative distribution function in (141). For each held-out observation, check the empirical distribution of

$$U_v^{(j)} = F_{v,C_e} \left(v^T C^{(j)} v \right). \quad (142)$$

Its correct marginal distribution is uniform on $[0, 1]$. Quantile plots, the maximum absolute difference from the uniform cumulative distribution, and errors in fixed tail probabilities provide conditional diagnostics. Serially correlated MCMC outputs are not independent uniform observations, so independence-based significance thresholds must not be used without a justified treatment of that correlation.

An orientation-sensitive RI check. For the RI example, write $C_e = U_e \Lambda_e U_e^T$, set $\tilde{C} = U_e^T C U_e$, and let $b_i = \kappa_0 c_0 + N \lambda_{e,i}$. Then $U_e^T B U_e = \text{diag}(b_i)$, and (139) implies, for $i \neq j$,

$$\mathbb{E}[\tilde{C}_{ij} | C_e] = 0, \quad \mathbb{E}[\tilde{C}_{ij}^2 | C_e] = \frac{b_i b_j}{(r+1)r(r-2)}, \quad \text{Var}(\tilde{C}_{ii} | C_e) = \frac{2b_i^2}{r^2(r-2)}. \quad (143)$$

Report the ratio of each sampled off-diagonal second moment to this positive exact value; the target ratio is one. Returning $U_e \Lambda U_e^T$ without orientation completion makes all these ratios zero, regardless of spectral accuracy. This deliberate negative control distinguishes matching a shrinkage estimate from recovering the full posterior.

Spectral summaries. Compare joint ordered log spectra and the conditional distributions of $\text{tr } C$, $\log \det C$, the largest eigenvalue, and the condition number $\lambda_{\max}/\lambda_{\min}$. The latter distributions can be compared with reference draws rather than approximated by functions of posterior mean eigenvalues. The log determinant also has analytic moments:

$$\begin{aligned} \mathbb{E}[\log \det C | C_e] &= \log \det B - d \log 2 - \sum_{i=1}^d \psi \left(\frac{n+1-i}{2} \right), \\ \text{Var}(\log \det C | C_e) &= \sum_{i=1}^d \psi_1 \left(\frac{n+1-i}{2} \right). \end{aligned} \quad (144)$$

Here ψ and ψ_1 are the first and second derivatives of $\log \Gamma$. These formulas follow from $\log \det C = \log \det B - \sum_i \log T_{ii}^2$ in (136) and the independent chi-squared diagonal variables. Spectral checks do not validate orientation; both kinds of diagnostics are necessary. Accuracy of arbitrary unbounded moments is not implied by a distributional error bound without suitable tail control.

H.4 MIXTURES WITH EXPLICIT POSTERiors AND AN EXACT SMALL-SCALE EM CONTROL

A finite mixture allows more varied posterior shapes while keeping the reference computable. Let p_h be the density of $\text{IW}_d(\nu_h, \Psi_h)$ and take

$$\pi_\star(dC) = \sum_{h=1}^H w_h p_h(C) dC, \quad w_h > 0, \quad \sum_h w_h = 1. \quad (145)$$

Applying (133) component by component gives

$$\begin{aligned} Q_\star(dC | C_e) &= \sum_{h=1}^H \tilde{w}_h(C_e) \text{IW}_d(\nu_h + N, \Psi_h + N C_e)(dC), \\ \tilde{w}_h(C_e) &= \frac{w_h z_h(C_e)}{\sum_\ell w_\ell z_\ell(C_e)}, \quad z_h(C_e) = \frac{|\Psi_h|^{\nu_h/2} \Gamma_d((\nu_h + N)/2)}{\Gamma_d(\nu_h/2) |\Psi_h + N C_e|^{(\nu_h + N)/2}}. \end{aligned} \quad (146)$$

The expression for z_h is the ratio of inverse-Wishart normalizing constants after updating; powers of two and observation factors common to all components cancel. Calculate weights from log determinants and log gamma functions, subtracting the largest log weight before exponentiation. Reference sampling first draws h with probability $\tilde{w}_h$ and then uses (136) for that component.

For an RI mixture, take $\Psi_h = \kappa_h c_h I_d$ and $\nu_h = d + 1 + \kappa_h$. For a non-RI mixture, use fixed nonspherical $C_{0,h}$ in place of $c_h I_d$. All relevant component moments must exist. Means and one-dimensional cumulative distributions are posterior-weighted averages of the component quantities. Variances include both within-component and between-component terms: if a_h and s_h^2 are the component mean and variance of $f(C)$, then

$$\mathbb{E}[f(C) | C_e] = \sum_h \tilde{w}_h a_h, \quad \text{Var}(f(C) | C_e) = \sum_h \tilde{w}_h (s_h^2 + a_h^2) - \left(\sum_h \tilde{w}_h a_h \right)^2. \quad (147)$$

Mixture uncertainty need not remain appreciable at every observation. Report accuracy separately on observations with uncertain component membership and those dominated by one component, using a grouping rule fixed before evaluating test performance (for example, whether the largest posterior component weight exceeds 0.9). Do not label generated matrices by a hard nearest-component rule. Instead calculate their exact component probabilities

$$a_h(C, C_e) = \frac{\tilde{w}_h(C_e) p_h^{\text{post}}(C | C_e)}{\sum_\ell \tilde{w}_\ell(C_e) p_\ell^{\text{post}}(C | C_e)}, \quad \mathbb{E}[a_h(C, C_e) | C_e] = \tilde{w}_h(C_e), \quad (148)$$

and compare the sampled average with the right-hand side. This avoids arbitrary labels for overlapping components. A mixture of isotropic-scale inverse-Wisharts still has a posterior mean of the form $a(C_e)C_e + b(C_e)I_d$. It tests distributional learning and data-dependent shrinkage, not general spectral shrinkage functions.

An exact empirical EM reference. There is a useful additional control for the SI-EM implementation. Start from any specified finite inverse-Wishart mixture and use a very small recorded ensemble. Its posterior at each observation is explicit by (146), so

$$\pi_{k+1} = \frac{1}{M} \sum_{a=1}^M Q_{\pi_k}(\cdot | C_{e,a}) \quad (149)$$

is again an exactly specified finite mixture. Its component count is at most M times the preceding count; no approximate component merging, pruning, or projection is permitted when calling this the exact reference. At small M and for a few iterations, compare the SI reconstruction distribution and its conditional posteriors with this reference at the same iteration. This isolates approximation of the EM step from the finite-data difference between EM and the true prior. The empirical log likelihood can also be checked for monotonicity, using (146) for the current mixture up to fixed observation terms. The exponential component growth makes this a diagnostic, not the scalable method. It tests empirical EM, not the population convergence theorem.

H.5 CONDITIONAL DISTRIBUTION COMPARISONS AND DECISION ACCURACY

Use independent training, validation, and test tasks. Keep the hidden test covariances for scoring and generate independent exact posterior draws at each test observation. Compute discrepancies conditional on each C_e before averaging across tasks; pooling generated matrices across observations could hide an algorithm that largely ignores its conditioning input. Report means, medians, and upper quantiles of the per-observation errors, together with variation across independent training seeds.

A joint distribution discrepancy. Supplement the analytic checks by maximum mean discrepancy (MMD), a difference between kernel averages from two distributions (Gretton et al., 2012). For two independent draws X, X' from the generated law and Z, Z' from the reference, define

$$\begin{aligned} \text{MMD}_h^2 &= \mathbb{E}k_h(X, X') + \mathbb{E}k_h(Z, Z') - 2\mathbb{E}k_h(X, Z), \\ k_h(A, B) &= \exp \left[-\frac{\|\log A - \log B\|_F^2}{2h^2} \right]. \end{aligned} \quad (150)$$

The Gaussian kernel in invertible log coordinates distinguishes probability distributions, but finite-sample power still depends on dimension and bandwidth. Estimate the kernel averages from samples, using the same documented bandwidth rule for all methods and an independent reference batch to set its scale. Repeat the calculation with joint ordered log-spectrum vectors to isolate spectral error. Specify whether diagonal terms are included in the sample estimator. An exact-reference versus exact-reference comparison at matching sample sizes quantifies its nonzero sampling fluctuation. Do not report formal independent-sample test thresholds for correlated MCMC output without a valid correction. The entire matrix law, not an estimated density in dimension $d(d+1)/2$, is compared this way.

Posterior intervals. For selected functions $f(C)$, compare central generated-posterior intervals at nominal levels 50%, 80%, and 95% with the exact conditional intervals. Across independently generated test tasks, also report the frequency with which the hidden $f(C)$ falls in the generated interval and the interval width. Use task-level uncertainty intervals for these coverage estimates. Simulation-based rank checks (Talts et al., 2018) are supplementary: some averages over observations drawn from the benchmark model can be correct even for a sampler that ignores C_e . The fixed-observation tests (142) and (137) are therefore primary here.

Connection to LW/LP. Report covariance-estimation risk separately from posterior accuracy. Under squared Frobenius loss, the exact benchmark action is $M_\star$, and for an estimate $A = A(C_e)$,

$$\mathbb{E}[\|A - C\|_F^2 \mid C_e] - \mathbb{E}[\|M_\star - C\|_F^2 \mid C_e] = \|A - M_\star\|_F^2. \quad (151)$$

The same identity holds after averaging independent Monte Carlo randomness used to construct A . Under the stated Stein convention, compare with $\Omega_\star^{-1}$; for mixtures, invert the posterior mixture mean of C^{-1} , not the individual component means before averaging. Include the empirical covariance, linear Ledoit–Wolf shrinkage, an implemented nonlinear shrinkage estimator associated with the LP oracle, and an inverse-Wishart prior fitted using only the same noisy ensemble. The last receives the same noisy data but assumes a restricted family; fit its parameters by maximizing the observed marginal likelihood obtained from z_1 in (146), not by using hidden clean covariances. The known-prior posterior and the supervised SI have additional information and are diagnostic references, not equally informed competitors. Neither LW nor nonlinear shrinkage supplies posterior intervals by itself. For empirical-PCA variance and covariance risk, compare the coefficients $\mathbb{E}[\xi_i \mid \Lambda_e]$, not $\mathbb{E}[\lambda_i \mid \Lambda_e]$.

H.6 SEPARATING ERRORS, CHOOSING TEST SIZES, AND REPORTING COST

A staged evaluation is more informative than testing only the completed algorithm. The following controls should use the same held-out observations and reference laws.

Reference and supervised checks. First verify the independent reference generator against (138)–(144), including a nonspherical scale to detect transposition errors. Then train the conditional SI on genuinely paired synthetic data in a separately labeled supervised experiment. Vary the number of SDE steps and compare the basic and Föllmer samplers with the same fitted drift. This tests the conditional sampler before introducing prior-learning error. Finally hide the clean targets and run the self-consistent algorithm, saving checkpoints across outer iterations. On a tiny dataset, use (149) to measure its error against exact empirical EM.

RI spectral and orientation checks. To test spectral learning, compare SI outputs with eigenvalues of independent exact posterior matrices. To test orientation alone, retain those exact reference eigenvalues, discard the reference eigenvectors, and complete each spectrum using the proposed orientation sampler. Correct completion must recover the known full posterior. Next combine learned spectra with orientation MCMC and compare the resulting matrices with the general matrix SI. Include $R = I_d$ as a deliberately incorrect completion and a sampler that always returns the posterior mean. These controls should fail the off-diagonal or variance checks even when some means or spectral summaries appear accurate.

For orientation MCMC, hold (Λ, Λ_e) fixed when studying mixing and use multiple initial orientations. Changing Λ changes the target: a sequence of chains at different spectra is not one stationary chain.

Stage	Starting sizes	Main comparison
Exact-step control	$d \in \{2, 4\}$, $N = 2d$, $M = 8$, up to three updates	Explicit mixture EM versus the corresponding SI update.
Posterior validation	$d \in \{4, 8, 16\}$, $N/d \in \{1, 2, 4\}$, initially $M = 5,000$	General and RI methods on the same RI tasks; general method on non-RI tasks.
Ensemble sensitivity	One validated (d, N) setting; $M \in \{1,000, 5,000, 20,000\}$	Effect of noisy-ensemble size, independent of other changes.
Scaling extension	$d \in \{32, 64\}$ and larger only as feasible, initially $N = 2d$	Complete reconstruction accuracy and cost, including orientation and spectral simulation.

Table 1: Proposed initial test settings. The exact-step control uses the same small recorded ensemble for the mathematical EM reference and its learned implementation.

One final draw from each independently initialized finite chain can be independent across spectra but still biased by insufficient run length. Conversely, repeated draws along a fixed-spectrum chain are correlated. Report initialization, discarded iterations, proposal types, run lengths, and estimated effective sample sizes for specified functions; effective sample size measures precision loss from serial correlation and is not a guarantee that transient bias has disappeared.

Suggested test sizes. Table 1 gives concrete starting settings to refine using validation data and the available computing budget. All values are proposed, not experimentally selected; the tests need not use every combination of settings. Use $c_0 = 1$ and initially $\kappa_0 = d + 8$; stress tests can use fixed $\kappa_0 = 8$ as d increases. For non-RI scales, try condition numbers 4 and 16 at mean eigenvalue one, keeping Q_0 fixed. For two-component RI mixtures, start with equal weights, $c_1 = 1 - \Delta$, $c_2 = 1 + \Delta$, and $\Delta \in \{0.15, 0.5\}$, rather than assuming both examples are equally difficult. Keep N fixed within each run unless the model explicitly conditions on it. For initial posterior validation, use separate validation and test sets of 256 tasks each, at least three independent training seeds, and $J = 512$ generated draws per selected test observation. Increase to $J = 2,048$ for detailed moment checks and use, for example, 8,192 independent reference draws when analytic quantities are unavailable. Use analytic moments whenever possible. Kernel comparisons may use fixed smaller subsamples to limit their quadratic cost. The reference sample size should itself be increased until reference uncertainty is small relative to the errors being compared.

Sweep SDE grids of 32, 64, and 128 steps before interpreting reconstruction differences as learning error. For orientation, increase the update budget geometrically and include multiple starts; no universal adequate run length is assumed. Compare interpolation scales $\sigma = a\sqrt{2/N}$ for a small declared set such as $a \in \{1/2, 1, 2\}$ while retaining $\sigma_t = \sigma(1 - t)$. Evaluate a fixed sequence of outer checkpoints rather than choose the best test-set iteration. Architecture and training budgets remain to be specified. Initialize from a declared sampleable prior not fitted using hidden clean data, and test sensitivity to its spread. The benchmark truth must not be supplied as the initialization merely to make the task easier.

What to plot and report. The first summary should compare posterior mean error with the RI off-diagonal second-moment ratios in (143); this shows whether the method goes beyond a shrinkage estimate. Additional panels should separate spectral-distribution error, orientation-completion error, and full-matrix error as their respective computational budgets increase. Include representative conditional directional-variance quantile plots, held-out coverage and width, and mixture-component probability errors. Score the same known-prior Bayes actions, LW/nonlinear shrinkage estimates, and learned actions against the same hidden test matrices. Report both average behavior and difficult observations, stratifying by empirical condition number using a rule fixed on validation data.

Report data-generation and eigendecomposition costs, network training, SDE evaluation, and orientation sampling separately, as well as end-to-end cost per full covariance draw. Use independent repetitions or correlation-aware estimates of Monte Carlo uncertainty. Record non-finite outputs and decoding failures rather than discarding or repairing them: clipping, trace normalization, or regularization changes the sampled law. Finite experiments assess practicality, not population convergence or uniform accuracy of all posterior moments.

I KNOWN COVARIATES: CONDITIONAL PRIORS, POSTERIOR, AND TESTS

This appendix extends the existing methods without changing their numerical state or interpolation schedule. A *covariate* is recorded information about a task, such as its experimental condition, which is available when its covariance is reconstructed. The mathematical statements below follow by conditioning the ideal results already proved in the paper. The conditional benchmarks and measured-data protocols are proposed additions, not completed experiments. Covariance regression models how covariance changes with explanatory variables (Hoff & Niu, 2012); here we instead learn a distribution of unobserved task covariances conditional on such variables.

I.1 CONDITIONAL MODEL AND THE TWO SAMPLING OBJECTIVES

Let c take values in a standard Borel space $\mathcal{X}$ (for example, a Euclidean space or a finite set), with population law η . Conditional distributions can then be chosen as measurable probability kernels. For fixed d, N , the model is

$$c \sim \eta, \quad C \mid c \sim \pi_\star(\cdot \mid c), \quad C_e \mid C, c \sim K_N(\cdot \mid C), \quad \mu_\star(\cdot \mid c) = K_N \pi_\star(\cdot \mid c). \quad (152)$$

The data are independent task pairs $(C_{e,a}, c_a)_{a=1}^M$. Neither C_a nor an evaluator or sampler for $\pi_\star(\cdot \mid c)$ is available. The forward assumption in (152) is $C_e \perp c \mid C$: once the covariance is supplied, the same Gaussian simulator applies. If the sampling design also varies, record it and use its known conditional channel $K(dC_e \mid C, c)$ instead. For example, a recorded sample size N_a changes the Wishart degrees of freedom and must enter both the simulator and the sampler. The results below use the fixed- N channel unless explicitly stated otherwise. Information computed from the same noisy samples, or used to select them, need not satisfy $C_e \perp c \mid C$; adding it as a label does not justify leaving the likelihood unchanged.

Write $Q_\star(dC \mid C_e, c)$ for the conditional posterior. At the population solution,

$$\pi_\star(dC \mid c) = \int Q_\star(dC \mid C_e, c) \mu_\star(dC_e \mid c). \quad (153)$$

This is the marginalization identity for the joint law $\pi_\star(dC \mid c) K_N(dC_e \mid C)$. It distinguishes the two useful outputs: a conditional SDE provides posterior draws at a supplied (C_e, c) , and averaging its exact draws over the conditional observation population provides conditional-prior draws. Sampling recorded pairs (C_e, c) and then drawing C approximates the joint population law of (C, c) ; discarding c gives the marginal prior. For finite observations or a fitted posterior, this is an approximation, not an independent source of exact clean data.

At a new continuous covariate value, evaluating a posterior network does not itself supply a draw from $\mu_\star(\cdot \mid c)$. Thus (153) is not a standalone generator of $\pi_\star(\cdot \mid c)$ at an arbitrary unobserved c . Such a generator would require conditional observation samples, or a separate conditional model trained on reconstructed (C, c) pairs, with its own approximation error. The prior-averaging claim in the title and main text does not assert that additional capability.

I.2 RETAINING THE COVARIATE IN THE TWO ALGORITHMS

Choose a sampleable initial family $\pi_0(\cdot \mid c)$; taking the same broad initial distribution for every c is permitted. At recorded or independently drawn design values of c , draw $C \sim \pi_0(\cdot \mid c)$, simulate C'_e using (2), and record (Y'_e, Y, c) . Use the matrix-log coordinates of Algorithm 1, or the log-spectral coordinates of Algorithm 2. In the latter case the initial family and the true conditional prior must be RI. The numerical state has the same dimension as before; c is an additional conditioning input, not a coordinate to be evolved.

At an iteration with current conditional prior π , the synthetic training law is $\eta(dc) \pi(dC \mid c) K_N(dC'_e \mid C)$. With the existing fixed positive scale σ , the interpolation and regression become

$$I_t = (1 - t)Y'_e + tY + \sigma(1 - t)W_t, \quad R_t = Y - Y'_e - \sigma W_t, \quad (154)$$

$$\mathcal{L}(\hat{b}) = \int_0^1 w(t) \mathbb{E} \|\hat{b}_t(I_t, Y'_e, c) - R_t\|_2^2 dt. \quad (155)$$

The expectation is over these triples and independent Brownian noise. The covariate is never interpolated. For a fixed supplied (C_e, c) , the Föllmer sampler is

$$dY_t = [(1+t)\hat{b}_t(Y_t, Y_e, c) - Y_t + Y_e]dt + \sigma\sqrt{1-t^2}dB_t, \quad Y_0 = Y_e. \quad (156)$$

The general decoder returns $C = \Phi^{-1}(Y_1)$. In the RI implementation it returns the population spectrum; orientation completion is described below. This is the same conditional-interpolant construction as in Proposition 1, with (Y_e, c) as the fixed conditioning variable (Chen et al., 2024).

Apply this sampler to recorded (C_e, c) , retain the reconstructed target *and the same* c , draw fresh Gaussian noise, calculate C'_e , and record (Y'_e, Y, c) for the next fit. In particular $C'_e \perp C_e \mid (C, c)$ under this construction. Shuffling c between targets, or replacing it after reconstruction, would change the synthetic joint law. Targets, covariates, and simulated observations are fixed during differentiation of the new regression; no derivative through the preceding SDE or simulator is needed. These are annotations to Algorithms 1 and 2, not a third learning algorithm.

I.3 IDEAL CONDITIONAL EM AND ITS GUARANTEES

For a conditional prior π_k , let $\mu_k(\cdot \mid c) = K_N \pi_k(\cdot \mid c)$ and let $Q_k(\cdot \mid C_e, c)$ be its exact posterior. Conditional on each c , the ideal update is

$$\pi_{k+1}(dC \mid c) = \int Q_k(dC \mid C_e, c) \mu_*(dC_e \mid c). \quad (157)$$

The distribution of c remains η throughout. This is EM for the joint model with a fixed covariate marginal and an unrestricted conditional mixing distribution. Re-corruption after (157) supplies $\eta(dc)\pi_{k+1}(dC \mid c)K_N(dC'_e \mid C)$, the training distribution for its next posterior sampler.

Define $D_k(c) = \text{KL}(\pi_*(\cdot \mid c) \parallel \pi_k(\cdot \mid c))$, $F_k(c) = \text{KL}(\mu_*(\cdot \mid c) \parallel \mu_k(\cdot \mid c))$, and $\bar{D}_k = \int D_k(c)\eta(dc)$, $\bar{F}_k = \int F_k(c)\eta(dc)$. Let $\bar{G}_k$ be the posterior KL averaged against $\eta(dc)\mu_*(dC_e \mid c)$.

Proposition 17 (Conditional ideal learning and recovery). *Assume $\bar{D}_0 < \infty$, and the exact regression, regularity, endpoint, and sampling hypotheses of Proposition 1 at every iteration for η -almost every c . Use the population observation law in reconstruction. Then (156) supplies the current conditional posterior; the output laws satisfy (157), and*

$$\bar{G}_k = \bar{D}_k - \bar{F}_k, \quad \bar{D}_{k+1} \leq \bar{D}_k - \bar{F}_k, \quad (158)$$

$$\bar{F}_k - \bar{F}_{k+1} \geq \int \text{KL}(\pi_{k+1}(\cdot \mid c) \parallel \pi_k(\cdot \mid c))\eta(dc), \quad \bar{F}_k \leq \frac{\bar{D}_0}{k+1}. \quad (159)$$

If $N \geq d$ and, for almost every c , the truth and initialization have support in a common compact set $K_c \subset \mathbb{S}_{++}^d$, then

$$\pi_k(\cdot \mid c) \Rightarrow \pi_*(\cdot \mid c), \quad Q_k(\cdot \mid C_e, c) \Rightarrow Q_*(\cdot \mid C_e, c) \quad (160)$$

for almost every (C_e, c) under the observed joint law. Alternatively, if $\bar{D}_k \leq \chi \bar{F}_k$ along the iterates for a fixed $\chi \geq 1$, then $\bar{D}_k \leq (1 - \chi^{-1})^k \bar{D}_0$ and the same upper bound holds for $\bar{G}_k$.

Proof. For fixed c , conditional expectation in (155) gives $b_t^\pi(y, Y_e, c) = \mathbb{E}[R_t \mid I_t = y, Y'_e = Y_e, c]$. The proof of Proposition 1 then applies conditional on c and gives the posterior endpoint. Averaging those endpoints yields (157). Finite $\bar{D}_0$ implies finite $D_0(c)$ for almost every c , so Theorem 3 applies to every such conditional problem. Its posterior identity follows from the KL chain rule for $\pi_*(dC \mid c)K_N(dC_e \mid C)$ and $\pi_k(dC \mid c)K_N(dC_e \mid C)$. Integrating that identity and the two descent inequalities proves (158)–(159); nonnegative integrands justify exchanging the order of integration and summation. In particular $\sum_k \bar{F}_k \leq \bar{D}_0$, and monotonicity gives the displayed rate. Corollary 4 applied at each c proves (160). Finally, the assumed averaged inverse-stability inequality turns (158) into the geometric contraction. None of these steps supplies that additional inequality or a useful value of χ . $\square$

Identifiability and finite covariate data. For fixed $N \geq d$, equality of the observed joint laws implies equality of their conditional observation laws for η -almost every c . Theorem 2 then identifies the corresponding conditional priors. For $N < d$, unrestricted conditional priors remain non-identifiable: choose the two indistinguishable priors in that theorem to be independent of c . Covariates can restrict a model class enough to help identification, but do not change this unrestricted counterexample. Likewise, the support and initialization qualifications of Section 3.4 hold separately at each c .

For a finite categorical covariate, empirical EM replaces each conditional observation law by the average within that group. With a continuously distributed c , the empirical joint update is

$$\hat{J}_{k+1}(dc, dC) = \frac{1}{M} \sum_{a=1}^M \delta_{c_a}(dc) \hat{Q}_k(dC | C_{e,a}, c_a). \quad (161)$$

It supplies reconstructed training triples, not an identified arbitrary conditional law at every possible c . A shared network imposes structure across covariate values; exact fitting at recorded inputs alone says nothing about unseen inputs. Repeated tasks, smoothness or other model restrictions, and covariate-level validation are therefore needed in finite data. Proposition 17 is an ideal population statement, not a theorem about that sharing mechanism or extrapolation. It also does not by itself ensure convergence of unbounded posterior moments.

I.4 CONDITIONAL ROTATIONAL INVARIANCE AND FULL-MATRIX COMPLETION

The relevant assumption is $C | c \stackrel{d}{=} Q C Q^T | c$ for every orthogonal Q , for η -almost every c . Let $\nu_k(d\Lambda | c)$ be the conditional prior on ordered population spectra. The spectral SI fits $q_k(d\Lambda | \Lambda_e, c)$ using the same d -dimensional state, the same spectral simulator (16), and the retained c . It does not fit independent scalar chi-square channels.

Proposition 18 (Conditional RI reduction and completion). *Assume conditional RI of the truth and initialization, and the unchanged Gaussian channel in (152). For almost every c , the RI reduction of Theorem 9 and its KL identities hold conditional on c . The full posterior is obtained by*

$$\Lambda \sim q_\star(\cdot | \Lambda_e, c), \quad R \sim H_N(\cdot | \Lambda, \Lambda_e), \quad C = U_e R \Lambda R^T U_e^T. \quad (162)$$

The orientation conditional H_N in (15) has no further dependence on c or $\nu_\star$. If a spectral approximation $\hat{q}(\cdot | \Lambda_e, c)$ is completed with this exact conditional, its matrix law $\hat{Q}^H$ satisfies, at fixed (C_e, c) ,

$$\text{KL}(Q_\star(\cdot | C_e, c) \| \hat{Q}^H(\cdot | C_e, c)) = \text{KL}(q_\star(\cdot | \Lambda_e, c) \| \hat{q}(\cdot | \Lambda_e, c)). \quad (163)$$

Under the ideal-learning assumptions, spectral EM therefore inherits Proposition 17, including its full-posterior conclusions after exact completion.

Proof. Condition on c . The prior representation is $\nu_\star(d\Lambda | c) dH(U)$. Inserting $C = U_e R \Lambda R^T U_e^T$ into the Gaussian likelihood yields the factor $|\Lambda|^{-N/2} \exp[-N \text{tr}(\Lambda^{-1} R^T \Lambda_e R)/2]$. Conditional on (Λ, Λ_e, c) , all dependence on the conditional spectral prior cancels, leaving exactly H_N . The proofs of Theorems 9 and 5 apply at this fixed c , including repeated eigenvalues when formulated for the matrix law. The same conditional matrix-given-spectrum law is shared by the exact and approximate completions, so its contribution to the KL chain rule is zero. Integrating the conditional spectral EM statements over η proves the final assertion. $\square$

A finite orientation run still contributes the error in Proposition 10, now evaluated at (Λ_e, c) . There is no orientation run during spectral training. If N is a recorded design covariate, its value must be used in H_N . Conditional RI leaves the learned state dimension d rather than $d(d+1)/2$, although simulation and orientation completion still involve matrix computations.

Conditional RI permits c to change the entire joint spectral law, but not to select preferred directions in the original variables. In particular, when its mean exists, $\mathbb{E}[C | c] = a(c)I_d$ for some scalar $a(c)$. This follows because the mean must commute with every orthogonal transformation. Thus known task labels do not automatically make a population RI. Predictable directional structure calls for the general method or an additional structural model.

I.5 AN OPTIONAL TRANSFORMATION FOR PREDICTABLE DIRECTIONAL STRUCTURE

One possible intermediate model is

$$C = B(c)^{1/2} \tilde{C} B(c)^{1/2}, \quad B(c) \succ 0 \text{ fixed and known}, \quad \tilde{C} \mid c \text{ RI.} \quad (164)$$

Here $B(c)$ supplies a background covariance pattern; RI is assumed only for the remaining random matrix. This is an additional model class, not a consequence of conditioning. Define $\tilde{C}_e = B(c)^{-1/2} C_e B(c)^{-1/2}$. Transforming each Gaussian observation gives

$$N\tilde{C}_e \mid \tilde{C}, c \sim W_d(N, \tilde{C}). \quad (165)$$

The conditional RI algorithm can therefore be applied to $\tilde{C}_e$, followed by decoding (164). Its learned variables are the spectra of the *transformed* matrices, not those of C and C_e . For fixed c , the congruence map in (164) is a bijection, so both posterior correctness and KL error transfer without loss to the original matrices. This follows by changing variables in both measures under the same bijection. The learned state is still d -dimensional; forming or applying a dense $B(c)^{\pm 1/2}$ adds computational cost.

A training-only estimate of the background is possible because $\mathbb{E}[C_e \mid c] = \mathbb{E}[C \mid c]$ when moments exist. For example, a regularized covariance-regression fit can be frozen before residual spectral training. Such fitting does not guarantee that the residual distribution is RI, and its error is not covered by the known- B assertion. Validation must test that assumption and compare with the background estimator alone. Keep this extension separate from the two main algorithms and do not estimate $B(c)$ using held-out clean covariances.

I.6 CONDITIONAL EXACT-POSTERIOR BENCHMARKS

Use the inverse-Wishart convention and sampling procedures of Appendix H. A conditional RI family is

$$C \mid c \sim \text{IW}_d(d + 1 + \kappa(c), \kappa(c)m(c)I_d), \quad m(c) > 0, \quad \kappa(c) > 6. \quad (166)$$

The concentration $\kappa(c)$ here is a prior parameter, not the multiplier used to tune interpolation noise. This lower bound supplies the fourth moments used for second-moment diagnostics. Conditional conjugacy (Alvarez et al., 2014) gives

$$C \mid C_e, c \sim \text{IW}_d(n(c), \Psi_e(c)), \quad (167)$$

$$n(c) = d + 1 + \kappa(c) + N, \quad \Psi_e(c) = \kappa(c)m(c)I_d + NC_e,$$

$$\mathbb{E}[C \mid C_e, c] = \frac{NC_e + \kappa(c)m(c)I_d}{N + \kappa(c)}. \quad (168)$$

The covariate changes both the prior mean scale and the strength of shrinkage. The precision mean, directional-variance distributions, off-diagonal second moments, log determinant, and transformed-Wishart checks in Appendix H.3 apply by substituting $n(c)$ and $\Psi_e(c)$ for n and B . Use independent reference draws at each fixed (C_e, c) , including for the joint spectral posterior. As before, inverse-Wishart support is not compact; Proposition 17’s compact-support conclusion does not automatically cover these benchmarks.

Categorical and continuous designs. For a first proposed test, use equally likely $c \in \{0, 1\}$, $(m(0), m(1)) = (1/2, 2)$, and $(\kappa(0), \kappa(1)) = (8, 32)$. For a continuous test, take $c \sim \text{Unif}[-1, 1]$, $m(c) = \exp(0.7c)$, and $\kappa(c) = 20 + 12c$. These are explicit design choices, not fitted results; the learner receives c but not these functions or the true prior parameters. Start with $d \in \{8, 16\}$, $N = 2d$, and the total training budget $M = 5,000$ of Table 1. Reuse that appendix’s validation splits, posterior-draw budgets, and noise/grid sweeps before increasing dimensions. Use a common, declared initialization that is not the conditional benchmark truth.

Compare a pooled learner that ignores c , one shared conditional learner, and, for categorical c , separate group learners. Use identical recorded ensembles and report both overall and per-group sample counts and training costs. Include a conditional inverse-Wishart fit from the same noisy observations, as well as the usual covariance-shrinkage baselines; the exact known-prior posterior is an information-privileged reference. The goal is to test the value of recorded information and

cross-group sharing, not assume that the conditional network must outperform every alternative. A null control keeps m, κ constant while supplying an independent covariate.

For the continuous design, report accuracy by predeclared covariate bins and on a fixed grid inside the training range. A separate held-out interval within that range tests interpolation across an unobserved region; values outside the training range test extrapolation and must be reported separately. Use the same conditional mean/precision errors, directional quantiles, coverage, widths, and matrix-distribution comparisons as in Appendix H, calculated for each test observation before aggregation. Average coverage alone can conceal overconfidence in one group and excessive width in another. Also compare covariance distributions reconstructed by averaging posterior draws within a categorical group with independent prior reference draws for that group. At continuous c , binning gives a distribution averaged over the bin, not an exact prior at its center.

Changing posterior shape and preferred directions. A conditional mixture retains an explicit reference posterior: take the two RI inverse-Wishart components from Appendix H.4, with component weight $w_1(c) = (1 + e^{-2c})^{-1}$ and $w_2(c) = 1 - w_1(c)$, and apply (146) at fixed c . This tests covariate-dependent component probabilities, not only a changing mean. A non-RI counterpart replaces $m(c)I_d$ in (166) by $B(c) \succ 0$. Its posterior is $IW_d(n(c), \kappa(c)B(c) + NC_e)$. For example, use $B(c) = Q(c) \text{diag}(b_1, \dots, b_d) Q(c)^T$ with fixed positive unequal b_i and a deterministic rotation of one coordinate plane by angle $\pi c/4$. The evaluator knows this function. It is not supplied to the general learner. Supplying $B(c)$ to the transformed-RI model is a separately labeled known-template control; estimating it from training observations is a different experiment. In this family the exact residual law is $IW_d(d + 1 + \kappa(c), \kappa(c)I_d)$, making both the transformation and full-posterior completion testable.

I.7 PROPOSED MEASURED-DATA APPLICATIONS AND VALIDATION

A suitable application provides many related tasks with limited observations per covariance and known information at prediction time. A large number of measurements from a single fixed covariance is not, by itself, an ensemble of covariance tasks. The following are candidate protocols, not evidence of measured-data accuracy. Covariance regression (Hoff & Niu, 2012) and population-based covariance estimation (Rahim et al., 2019) are relevant comparisons because they also share information across tasks.

EEG with known experimental condition. The PhysioNet EEG Motor Movement/Imagery data contain 64-channel recordings with baseline, executed-movement, and imagined-movement runs (Schalk, 2009). A first protocol would treat a participant-condition recording as a task, use a short approximately stationary segment to calculate C_e , and retain the known condition as c . Fix electrode selection and preprocessing using training participants only, and hold out participants rather than split nearby windows randomly. Compare pooled, condition-specific, and shared conditional reconstruction. Repeated tasks within a participant may be dependent; they do not automatically satisfy the independent-task population assumptions. Condition labels may be inputs when the task is covariance reconstruction under a known condition; they must not be supplied when evaluating prediction of that same unknown label. Assess predictive directional variances and uncertainty using independent or adequately separated validation measurements from the same condition, subject to the common-covariance assumption below.

Regional fMRI with participant or acquisition information. Covariate-assisted covariance analysis has been applied to functional magnetic resonance imaging from the Human Connectome Project (Park, 2025). Fix a modest common set of brain regions, define a task as a participant within a specified acquisition condition, and use only covariates available when reconstructing that participant’s covariance. Split participants, and related participants where relevant, before fitting any population model; never fit the template on validation scans. Compare with population shrinkage such as PoSCE (Rahim et al., 2019) and a conditional covariance-regression baseline. Named regions have directional meaning, so strict conditional RI is a restricted comparison, not a realistic assumption established by the dataset. The general model can learn these directions. The optional transformed-RI model can test whether a training-estimated background captures enough of them to make residual spectral learning useful. The latter assumption must be checked rather than inferred from successful mean normalization.

Validation through future noisy observations. A long recording gives a less noisy estimate, not the true covariance. If training and validation recordings are conditionally independent given the same (C, c) , the relevant predictive law is

$$P(dC_e^{\text{val}} | C_e^{\text{train}}, c) = \int K_{N_{\text{val}}}(dC_e^{\text{val}} | C) Q(dC | C_e^{\text{train}}, c). \quad (169)$$

Generate this law by drawing a posterior covariance and then a new validation-sized observation. Compare observable directional variances, spectral summaries, and interval coverage across held-out tasks and covariate groups, including both uncertainty in C and new sampling noise. If the covariance changes between recordings, the common- C assumption fails and a discrepancy need not be a posterior-sampling error. Likewise, EEG and fMRI samples are time-indexed: independent Gaussian sampling with N equal to the number of time points cannot be assumed without assessment. Subsampling, filtering, or replacing N by an effective sample size is at most an approximation unless justified. Either validate a suitable preprocessing/model combination or specify a known forward simulator that includes temporal dependence; the Wishart-specific theorems then apply only to the original model. Estimating and subtracting a mean also changes the degrees of freedom in the independent Gaussian case.

Other settings and the role of RI. Covariance estimation across physical simulation parameters is another natural conditional application; cosmology-dependent covariance modeling already uses simulations across parameter settings (Morrison & Schneider, 2013). Here c should include known physical and observing parameters, while any remaining task variation giving a nontrivial $\pi(C | c)$ must be specified. If c determines C exactly, the conditional prior is a point mass and the smooth posterior-sampler assumptions need separate treatment. Antenna-array covariances could similarly use known location or operating conditions, but their complex-valued observation model and stationarity units must first be specified. Neither example is claimed as a ready-to-run benchmark in this draft. For larger dimensions, prioritize the conditional RI implementation when that assumption is appropriate, or explicitly test the transformed-RI model. Report matrix preprocessing, spectral simulation, and orientation costs as well as neural-network cost: a smaller learned state does not make all computations linear in d .